

MATHEMATICS GRADE 10: SIMILARITY AND ENLARGEMENT

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.1 (12 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Mathematics
Strand	Strand 2.0: Measurements and Geometry
Sub-Strand	Sub-Strand 2.1: Similarity and Enlargement
Total Duration	12 lessons × 40 minutes = 480 minutes total
Sub-Strand Content	– Similar figures and their properties – Linear scale factor (LSF) – Area scale factor (ASF) – Volume scale factor (VSF) – Enlargement: centre of enlargement and scale factor (positive, fractional and negative) – Constructing images under enlargement – Determining the centre and scale factor of an enlargement – Application of similarity and enlargement in real life (maps, scale drawings, photographs, models)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - identify similar objects in the environment, - state the properties of similar figures, - determine the linear scale factor of similar figures, - determine the area scale factor of similar figures, - determine the volume scale factor of similar figures, - construct images under enlargement given the centre and scale factor, - determine the centre and scale factor of an enlargement, - apply similarity and enlargement in real life situations, - appreciate the use of similarity and enlargement in day-to-day life.
Core Competencies	– Critical Thinking and Problem Solving: the learner works out scale factors and uses them to solve problems involving lengths, areas and volumes. – Communication and Collaboration: the learner discusses with peers the properties of similar figures and justifies enlargement choices. – Creativity and Imagination: the learner draws and designs enlarged and reduced images of figures. – Digital Literacy: the learner uses digital tools such as GeoGebra or photo editors to explore enlargement.
Core Values	– Responsibility: the learner takes care of geometrical instruments and shares tasks fairly in group work. – Unity: the learner works with peers to investigate similar figures and to build shared explanations.

	– Integrity: the learner records measurements honestly and reports results accurately.
Science & Engineering Practices	– Asking questions and defining problems: students ask why a passport photo keeps its shape when resized. – Developing and using models: students build and revise a scale-factor model of enlargement. – Analyzing and interpreting data: students compare measured lengths and areas of original and enlarged figures. – Using mathematics and computational thinking: students use LSF, ASF and VSF to predict and calculate. – Constructing explanations: students justify why shape is preserved while size changes.
Pertinent & Contemporary Issues (PCIs)	– Life Skills (critical thinking, self-esteem): the learner participates confidently in discussions and defends reasoning about scale factors. – Financial literacy: the learner uses area and volume scale factors to compare costs of photo prints, paint and containers of different sizes.
Career Connections	– Graphic designer / photographer: resizing images and prints using scale factors. – Architect and civil engineer: scale drawings and building models. – Surveyor and cartographer: maps of Nairobi, Mombasa and Kisumu drawn to scale. – Tailor and fashion designer: scaling patterns for different sizes. – Printer or signwriter: enlarging posters and billboards.
Focus for Lessons	This unit uses the everyday puzzle of resizing a passport photo to develop the ideas of similarity, scale factor and enlargement. Learners predict, measure, explain and model how lengths, areas and volumes change under enlargement. They then apply the ideas to maps, scale drawings and Kenyan real-life contexts.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can a passport photo be made bigger or smaller without changing its shape, and how do its length, area and cost change when we do so? KICD KEY INQUIRY QUESTIONS: 1. How do we apply similarity and enlargement in day-to-day life? 2. Where else in our environment do we see objects that are similar?
Anchoring Phenomenon	A student in Kisumu needs passport photos for a KCSE registration and a school ID. The photographer prints the same photo in a small stamp size and in a larger postcard size. Both pictures look the same shape and nothing seems stretched, yet every measurement on the picture has changed. When the photo is made twice as wide, the paper needed does not simply double, and the shop charges more than twice. Passport photo resizing is a mathematical enlargement using a specific linear scale factor (LSF). It is puzzling because the shape stays identical while lengths, area and cost change differently, and students must work out the rule that connects them.
Storyline Thread	Lesson 1: Students predict and measure two photo sizes of the same passport picture and notice same shape but different size; the Driving Question Board (DQB) is started. Lesson 2: Identifying similar objects in the environment (photos, maps, matatu models, flags) and sorting them as similar or not similar. Lesson 3: Properties of similar figures: corresponding angles are equal and corresponding sides are proportional; testing with triangles and rectangles. Lesson 4: Determining the linear scale factor (LSF) from the original and the resized photo; model first draft. Lesson 5: Using LSF to find unknown lengths in similar figures, including the shadow of a tree in Kericho. Lesson 6: Area scale factor: predicting and investigating on grid paper that $ASF = (LSF)^2$; explaining why the photo paper cost is

	more than doubled. Lesson 7: Volume scale factor: building models with cubes and comparing jerrycans to show $VSF = (LSF)^3$. Lesson 8: Enlargement: locating the centre of enlargement and constructing images with a positive scale factor. Lesson 9: Enlargement with fractional and negative scale factors; reductions and inverted images. Lesson 10: Finding the centre and scale factor of an enlargement from an object and its image by drawing lines. Lesson 11: Applying similarity and enlargement to maps, scale drawings and costs in Nairobi, Mombasa and Kisumu contexts. Lesson 12: Consolidation: students finalise their enlargement model, answer the DQB and explain the passport photo resizing using LSF, ASF and VSF.
Supporting Phenomena	– A map of Kenya drawn on a 1:5,000,000 scale shows the distance from Nairobi to Mombasa as only a few centimetres. – A tailor in Gikomba makes a small and a large school uniform from the same pattern, and the larger one needs far more than twice the cloth. – A 500 ml and a 1 litre jerrycan of cooking oil look similar, but the 1 litre one is not twice as tall. – A shadow of a tall tree at a Kericho tea estate and a metre stick at the same time form similar triangles, letting us find the tree height without climbing.

ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (40 minutes (1 lesson)): Same Photo, Two Sizes: Launching the Driving Question Board

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon: Achieng's passport photo printed as stamp size S and postcard size P. Learners notice that the shape is the same but every measurement is different, predict how length, area and cost change, and start the Driving Question Board (DQB).
Knowledge	Learners observe that prints S (3 cm x 4 cm, KES 30) and P (6 cm x 8 cm, KES 120) show the same picture in the same shape but in different sizes. They record that lengths such as width, height, face height and diagonal are different on each print, and that the price is different. They do not yet have a rule connecting the changes.
Skills	Measuring lengths accurately in cm using a ruler; recording data in a table; making and justifying predictions; asking investigable questions (NGSS SEP: Asking Questions; Planning and Carrying Out Investigations; Analyzing and Interpreting Data); communicating and collaborating in groups.
Attitudes	Curiosity about everyday mathematics in a Kenyan setting such as a photo shop in Kisumu; appreciating that numbers can explain what we see; unity and respect when sharing ideas in groups; responsibility in handling rulers and printed photos.
Key Inquiry Question	How do we apply similarity and enlargement in day-to-day life? Where else in our environment do we see objects that are similar?
Purpose in Storyline	This is the opening lesson. It builds on no earlier lesson. Learners meet Achieng, Mr Otieno and the two prints, collect first measurements, make predictions about length, area and cost, and generate the questions that the DQB will track. Later lessons will answer these questions step by step.
Safety Notes	Use rulers safely and do not point them at classmates. Keep the classroom floor clear of loose papers. Use only paper prints and ordinary rulers. No sharp tools are needed; if cutting is required, the teacher cuts the prints beforehand.

B. LESSON OVERVIEW

Ms Wanjiru tells the story of Achieng, a Grade 10 student in Kisumu who needs passport photos for KCSE registration and a school ID. Mr Otieno, the photographer, prints the same picture as stamp size S (3 cm x 4 cm, KES 30) and postcard size P (6 cm x 8 cm, KES 120). Both look the same shape, nothing seems stretched, yet every measurement has changed, and the price more than doubled when the photo was made twice as wide. Learners first predict what will change and what will stay the same, then measure the width, height, face height and diagonal of prints S and P (in cm) and record the price. They explain what they see in pairs and groups, and then build the first Driving Question Board with the driving question and their own questions. Finally they sketch a first-draft idea of why the photo changes in size but not in shape. The lesson only launches the puzzle: it names the idea of same shape but different size, and it does not teach how to calculate scale factors, area factors or cost rules, which come in later lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners listen to the story of Achieng at Mr Otieno's photo shop	Say: 'Imagine you are Achieng in Kisumu. You need photos for	Predict-Observe-Explain (prediction stage) with a Think-	Look for: (1) learners who write a price and a reason on the Predict

 VIDEO: Dilations: scale factor Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	in Kisumu. Ms Wanjiru shows (or draws on the board) the stamp print S and the postcard print P without revealing any measurements. Individually, then in pairs, learners complete a Predict sheet: (1) Will the shape of the picture be the same or different on the two prints? (2) Which measurements will change: width, height, face height, diagonal? (3) Mr Otieno charges KES 30 for S and the postcard P is made twice as wide: what do you think he will charge for P? Why? Learners write a number and a reason, and draw a quick sketch of what they think P looks like next to S.	KCSE registration and a school ID. Mr Otieno prints the same photo small and big.' Show the two prints for 10 seconds, then hide them. Ask: 'What do you think will be the same, and what will be different?' Give WAIT TIME of 10-15 seconds in silence before anyone speaks. Ask learners to write their prediction alone for 3 minutes. Then cold-call 4 learners from different parts of the room to share the price prediction and reason, writing each prediction on the board without judging (for example 'KES 60' and 'KES 120'). Say: 'Thank you. We will keep these predictions and test them. Nobody is right or wrong yet.' Walk around and note learners who predict 'double the price' since this is the common idea to revisit in later lessons.	Pair-Share. Writing a prediction with a reason before seeing data makes learners' existing ideas about size, shape and price visible, so they have something to compare with the evidence later. Sharing in pairs lets them test ideas safely before the whole class.	sheet (target: at least 80% of learners); (2) the number of learners who predict that the price doubles from KES 30 to KES 60 (record the count on the board); (3) whether learners use words such as 'bigger', 'same shape' or 'stretched' to describe the picture; (4) whether sketches show P as a scaled copy of S or as a stretched picture. Record these ideas to revisit in Lessons 4 and 6.
Observe Phase  VIDEO: Side lengths after dilation Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations and properties Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4, learners receive printed or drawn copies of Print S (3 cm x 4 cm) and Print P (6 cm x 8 cm) of the same passport picture. They use a ruler to measure in cm: width, height, face height and diagonal for each print, and record these in a table titled 'Achieng's Prints'. They also copy the price from Mr Otieno's price list (S: KES 30, P: KES 120). Learners then compare the two prints by placing S on top of P at one corner and describe what they see. They check whether the corners look the same (all right angles, same face shape) and whether the picture looks stretched. Each group writes at least three observations of what is the same and three of what is different.	Hand out materials and say: 'Measure to the nearest half a millimetre and write units in every cell.' Circulate and ask: 'Which side of S matches which side of P?' and 'How do you know your measurement is accurate?' Check that learners measure from the 0 mark, not the ruler's edge. Give WAIT TIME of 10-15 seconds after each question. Call 'Pencils down' after 12 minutes. Cold-call 3 groups to report one measurement each from the table (for example S width 3 cm, P width 6 cm, S face height 2.4 cm, P face height 4.8 cm, S diagonal 5 cm, P diagonal 10 cm) and fill a class table on the board. Ask: 'What do you notice about the numbers in the two rows?' Do not name the rule; accept learner words such as 'it looks like double' and record them as ideas to test. Mention the price	Data table and noticing and wondering. A structured table of measurements lets learners see patterns across both prints. 'I notice... I wonder...' sentence frames encourage learners to describe evidence first and turn puzzles into questions, before any formal rule is given.	Observe: (1) measurements within 1 mm of 3 cm x 4 cm and 6 cm x 8 cm (target: most groups); (2) units written in every cell; (3) groups that match corresponding sides (width with width, height with height, face height with face height); (4) at least three 'same' and three 'different' observations. Prompt: 'Which measurements changed, and did the shape change?' Note groups that already spot that the lengths in P are twice those in S but not the price, for use in the next phase.

		for the first time: 'S costs KES 30 and P costs KES 120. Is that what you predicted?'		
Explain Phase  VIDEO: Dilations: scale factor Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups use their table to explain the puzzle in their own words. They complete these sentence frames: 'The two prints have the same ____ but different ____.' 'Every length on P is ____ compared with S.' 'The price changed from ____ to ____, which is ____ than I predicted.' Each group then compares their original prediction to the measured data and underlines what they got right and what surprised them. Learners discuss the question: 'If Mr Otieno made a print twice as wide, why is it not simply twice the price?' They are allowed to suggest ideas such as 'more paper is used' but are not expected to give a full rule. Groups write one claim, one piece of evidence from their table, and one remaining question on a half sheet of paper.	Say: 'Use the numbers in your table as evidence. Do not just tell me what you think; show me which numbers.' Give WAIT TIME of 10-15 seconds after asking: 'Why might the price be more than double when the width is double?' Cold-call 4 learners (from at least 3 groups) to share claims and pick up on disagreements by asking: 'Who sees it differently, and what evidence do you have?' Introduce vocabulary only as a label: 'When a picture changes size but keeps its shape, we say the two pictures are similar. The original is the object and the bigger picture is the image.' Say that in this unit they will find out how to describe and calculate the change. Do not teach scale factor calculations, area factors or cost rules today; if a learner says 'the lengths double' say: 'Good observation. We will test and name that idea in later lessons.' Keep the Kenyan context by linking to Kisumu, KCSE registration and the school ID.	Claim-Evidence-Reasoning with a class discussion (Talk Moves: revoicing, adding on, agree/disagree). Linking each claim to numbers from the table trains learners to justify ideas with data, and discussion of conflicting ideas, such as the price prediction, creates a real need for the mathematics in later lessons.	Listen for: (1) learners who use evidence such as '3 became 6 and 4 became 8' to support their claim; (2) correct use of 'same shape' versus 'same size'; (3) learners who recognise that the price is not simply double the length and say why they think so; (4) correct use of the words object, image and similar. Collect the half sheets and sort them into: has claim and evidence; has claim only; confused about shape and size. Use the sort to plan support in Lesson 2.
Driving Question Board (DQB) Creation  VIDEO: Side lengths after dilation Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations and properties	Ms Wanjiru reveals the Driving Question Board, a large chart paper on the wall with the driving question at the top: 'How can a passport photo be made bigger or smaller without changing its shape, and how do its length, area and cost change when we do so?' Beside it is a photo of prints S and P. Each learner writes at least one question on a sticky note (for example 'Why did the price become four times, not two?' or 'Is it always the same shape when you make a photo bigger?' or	Say: 'A good question helps us investigate. Write the questions you really wonder about.' Model one example: 'How many times bigger is the paper in P than in S?' Give WAIT TIME of 10-15 seconds so that every learner can think of at least one question. Ask 5 learners (cold-call) to read out and place their notes; ask 'Which group has a similar question?' to cluster notes. Show the KICD key inquiry questions as extra prompts if the board is thin: 'How do we apply similarity and enlargement in day-	Driving Question Board with question clustering. Learners' own questions give ownership of the learning path, show how the phenomenon will be investigated across the unit, and help the teacher link future lessons to their curiosity, with the class coherently working towards answering the driving question.	Check: (1) every learner has written at least one question on a sticky note; (2) at least some questions refer to length, area or cost, not only to shape; (3) learners can explain how their question links to the photo of S and P; (4) groups can sort the notes into at least three categories. Make a quick tally of question types (shape, length, area, cost, real life) to see what to emphasise in later lessons.

Source: Khan Academy (English - US curriculum) Search ARES for similar readings	'Where else do we see bigger and smaller copies, like maps or a model matatu?'). Groups then sort their notes into categories: shape, length, area, cost, and real life uses. A few learners place their questions on the DQB and read them aloud. The class agrees on three questions they most want to answer.	to-day life?' and 'Where else in our environment do we see objects that are similar?' Add them to the board. Say: 'We will come back to this board in every lesson, move questions we can answer, and add new ones.' Photograph or keep the board safe for later lessons. Make sure all questions are respected, including ones that are not mathematical.		
Model Building Phase  VIDEO: Dilations: center <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Dilations: center <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Each learner starts a first-draft model in their notebook titled 'How Achieng's photo changes size'. They draw print S (3 cm x 4 cm) and print P (6 cm x 8 cm) side by side, label the width, height and face height of each, and add arrows to show what is changing. Beneath the drawings, they write: (1) what stays the same (the shape, the picture, the corners), (2) what changes (the lengths, the paper used, the price), and (3) one 'I think the rule might be...' statement and a sentence about what they still need to find out. Learners swap models with a partner and add one comment or question on a sticky note. They then keep the model to revise in later lessons.	Say: 'This is a first draft. It is allowed to be incomplete. Show what you think is happening and where you are unsure.' Give WAIT TIME of 10-15 seconds before learners begin drawing. Circulate and ask: 'What does the arrow show?' and 'What evidence from your table supports that?' Cold-call 3 learners to share their 'I think the rule might be...' statement and write the ideas on a chart called 'Our first ideas'. Close with the summary: 'Today we noticed that the picture has the same shape in both sizes but all measurements and the price are different. In the coming lessons we will find out how to describe this with numbers.' Ask learners to look for two objects at home, such as a map, a photo or a model matatu, that are the same shape but different size, and bring a description for Lesson 2. Return the DQB to display. Have learners complete the summary table (observed, learned, explained).	Initial model building with peer feedback. Drawing and labelling the prints and writing a tentative rule makes thinking visible, gives a baseline model that learners will revise as they gain evidence, and peer comments model the habit of scientific critique.	Check models for: (1) both prints drawn with correct labelled dimensions (3 x 4 cm and 6 x 8 cm); (2) at least one thing that stays the same and one that changes listed correctly; (3) a tentative rule or question about the price or lengths; (4) a peer comment given. Success is at least 80% of learners with a complete first draft. Note learners who think the shape changes or that price follows length, and plan to address these in Lessons 4 and 6.

D. TEACHER REFLECTION

Which predictions about the price (for example KES 60 versus KES 120) did learners make, and how many held on to the idea that doubling the width doubles the price? Did learners measure correspondingly matched sides accurately and confidently? Did every learner contribute at least one question to the DQB, and which question types were missing? Which learners confused 'same shape' with 'same size', and what support will they need in Lesson 2? How well did the Kisumu photo-shop context

engage learners, and what other local examples (maps, a model matatu) could be used next?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We saw two prints of Achieng's passport photo: stamp size S (3 cm x 4 cm, KES 30) and postcard size P (6 cm x 8 cm, KES 120). The picture has the same shape on both prints, but the width, height, face height, diagonal and price are all different.
What did I learn?	A picture can be made bigger without changing its shape, and we call such pictures similar. The original is the object and the bigger copy is the image. Different measurements change in different ways, and the price did not simply double.
How does this explain the phenomenon?	We think the lengths on P are bigger than on S because the photo was enlarged, and the price is more than double because more paper is used, but we do not yet know the rule. We have started the DQB with questions about length, area and cost to investigate in later lessons.

LESSON 2 (40 minutes): Spotting Similar Objects Around Us

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners identify and sort objects in their environment (photos, maps, flags, model matatu) as similar or not similar by looking at shape, using Achieng's passport photos S and P and the distorted 6 x 4 photo as anchor evidence. This lesson uses visual judgement only. Testing similarity with numbers (angles and side ratios) belongs to Lesson 3.
Knowledge	Similar objects have the same shape but may differ in size. Photos S (3 x 4 cm) and P (6 x 8 cm) are similar. The distorted 6 x 4 photo, made by doubling only the width, is stretched and is not similar to S. A model matatu at 1:25 is similar to the real matatu (700 cm real, 28 cm model). Use 'similar' for same shape and 'congruent' only for same shape and size.
Skills	Observing and comparing shapes carefully (NGSS Practice 4: Analysing and Interpreting Data). Sorting and classifying objects using a stated reason (Practice 3: Planning and Carrying Out Investigations). Constructing an explanation using evidence (Practice 6). Communication and Collaboration: discussing with peers to sort objects. Critical Thinking and Problem Solving: justifying why objects are or are not similar.
Attitudes	Unity: working in groups to agree on and justify a sort. Curiosity about similar objects in the environment. Appreciating the use of similarity and enlargement in day-to-day life. Life Skills (self-esteem): participating confidently in group discussion and defending a reasoned choice.
Key Inquiry Question	How do we apply similarity and enlargement in day-to-day life? Where else in our environment do we see objects that are similar?
Purpose in Storyline	Lesson 1 showed that Achieng's prints S and P have the same shape but different size and launched the DQB. Lesson 2 builds the first piece of evidence for the driving question: what 'same shape' means in everyday objects and how a distorted photo differs from an enlargement. This gives learners a shared vocabulary and a sorting rule to test with numbers in Lesson 3.
Safety Notes	Use paper cut-outs and printed cards only. If using scissors to cut cards, cut away from the body and keep them on the desk. Keep bags and walkways clear during gallery walk movement. Handle any model matatu gently.

B. LESSON OVERVIEW

Ms Wanjiru begins by showing Achieng's photo S (3 x 4 cm) beside P (6 x 8 cm) and the distorted 6 x 4 photo. Learners predict which pictures are the same shape as S. They then sort a set of cards in groups: photos S, P and the distorted photo, a Kenyan flag and a smaller copy, a Kenyan flag stretched wide, a Nairobi-Mombasa map and a smaller copy, a real matatu (7 m = 700 cm) and a 1:25 model (28 cm), and a square versus a rectangle. Groups give a reason for each decision and walk the gallery to challenge others. In the Explain phase learners agree a class sorting rule using 'looks the same shape, only the size changes' and find where it fails (stretching). They add questions to the DQB and add a first sentence to the draft model: similar objects keep their shape, and the distorted photo does not.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Dilations and	Learners recall Achieng at Mr Otieno's shop in Kisumu. They see photos S (3 x 4 cm), P (6 x 8 cm)	Display the three pictures. Say: 'Achieng asked Mr Otieno for another print. He made P by	Predict-Share-Revisit: learners commit to a prediction and a reason in writing before seeing	Look for learners who write a reason that refers to shape (for example 'the face looks fatter', 'it is

properties Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Dilations and properties Source: Khan Academy (English - US curriculum) Search ARES for similar readings	and the distorted photo (6 x 4 cm, width doubled only) drawn on the board or printed cards. Individually, they write a prediction in their books: 'Which of these photos has the same shape as S? Which looks stretched? Why?' They then share with a partner.	doubling the width and the height. Another print was made by doubling the width only. Which one would Achieng be happy to put on her KCSE registration form?' Give WAIT TIME of 10-15 seconds in silence before anyone writes. Walk around and note predictions without correcting. Cold-call 3 learners to read their predictions and reasons. Ask: 'What word would you use to describe the shape of S and P?' Record learner words on the board (same shape, not stretched, bigger) without confirming them yet.	evidence. This surfaces everyday ideas about 'same shape' and creates a reason to test them. The predictions are revisited in the Explain phase.	just bigger') and not only size. Tally on a sheet how many choose S and P as the same shape. Note learners who think the bigger picture is a different shape.
Observe Phase VIDEO: Similar shapes & transformations Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Dilations and properties Source: Khan Academy (English - US curriculum) Search ARES for similar readings	In groups of 4, learners receive a card set: (1) S and P; (2) S and the distorted 6 x 4 photo; (3) a Kenyan flag and a smaller copy; (4) a Kenyan flag and a stretched-wide copy; (5) a map of Kenya and a smaller copy; (6) a real matatu 7 m long (700 cm) and a model matatu 1:25 (28 cm long); (7) a square and a rectangle. Where cards are not available, learners draw the pairs from the board. They sort each pair into 'similar' or 'not similar', write the reason, and then do a gallery walk to compare their sort with other groups and use sticky notes for disagreements. Learners also list two other similar objects they see in the classroom or school compound, for example a small and large exercise book or a school badge on a sweater and on a notice board.	Say: 'Work as a team. Every pair needs a decision and a reason that uses the word shape.' Circulate and ask: 'If I shrink this picture on a phone, what must happen to both width and height for it to look the same?' and 'Is the stretched flag the same flag? Is its shape the same?' Give WAIT TIME of 10-15 seconds after each question before taking answers. Prompt groups that sort by size only: 'Does big or small decide similar? Look at S and P again.' During the gallery walk, ask each group to leave 2 sticky-note questions. Tell learners that numbers will be used to check their decisions in Lesson 3, so today they use careful looking. Cold-call 4 groups to report the pair they found hardest to decide (square and rectangle is the expected one).	Sorting and Classifying with Gallery Walk: learners build a category from examples and non-examples, then compare decisions across groups. Seeing contrasting pairs (S and P versus S and the distorted photo) makes the key feature visible: similar means same shape even when size changes.	Check that groups correctly sort S and P, flag and smaller flag, map and smaller map, and real matatu and 1:25 model as similar, and the distorted photo and stretched flag as not similar. Listen for reasons like 'only the width changed'. Note any group that sorts by size alone. Collect sorting sheets as evidence.
Explain Phase VIDEO: Dilations and properties Source: Khan Academy (English)	Learners return to their predictions from the Predict Phase and revise them. As a class, they discuss why S and P are similar but S and the distorted photo are not, and agree on a class sorting rule in their own	Say: 'Look back at your predictions. What would you keep and what would you change?' Give WAIT TIME of 10-15 seconds. Ask: 'S and P differ in size and the distorted photo differs in size too.	Claim-Evidence-Reasoning with Talk Moves: learners state a claim (for example 'P is similar to S'), support it with observed evidence from the cards, and give a reason linked to shape. Talk moves (add	Check written sentences for the idea that similar means same shape with possibly different size, and for correct use of 'congruent'. Look for learners who explain the distorted photo as 'width changed

- US curriculum) Search ARES for similar videos EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	words. They test the rule on the model matatu: the model is much smaller than the real matatu (28 cm against 700 cm) but still the same shape, so it is similar. They also say why the distorted photo would look wrong on Achieng's school ID. Each learner writes one sentence beginning 'Two objects are similar when...' and one example from Kenya.	So what makes S and P similar but not the distorted photo?' Cold-call 3 learners, then ask 2 others to add on or disagree using 'I agree because...' or 'I disagree because...'. Point to the matatu: 'A real matatu is 700 cm and the model is 28 cm. Is the model similar? What stays the same?' Write the agreed class rule on chart paper: 'Similar objects have the same shape; only the size can be different.' Do NOT teach angle or ratio tests, but say: 'In Lesson 3 we will check this rule with numbers.' Also introduce 'congruent' (same shape and same size) briefly and contrast it with 'similar'.	on, agree or disagree with a reason) make reasoning public and let learners refine the class rule together.	but height did not'. Ask 3 learners to give their own Kenyan example and check that it is genuinely similar (for example a small and large Kenyan flag).
Driving Question Board (DQB) Creation VIDEO: Similar shapes & transformations Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Similarity & transformations <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners add sticky-note questions to the DQB under the driving question about the passport photo. They use today's evidence and their gallery walk questions. They sort their notes into themes such as 'How do we tell if shapes are similar?', 'What happens to length?', 'What happens to area and cost?' and 'Where else is similarity used?'. They place notes next to the Lesson 1 questions and mark any question partly answered today.	Say: 'Write one question that today's sorting made you wonder about, for example how we can be sure two shapes are similar, or why the price jumps.' Give WAIT TIME of 10-15 seconds, then allow 2 minutes for writing. Cold-call 4 learners to read and place their questions. Ask: 'Which earlier question can we now partly answer?' Move the question 'What does same shape mean?' to a 'Partly answered' area. Keep questions about area and cost for later lessons. Link to the KICD key inquiry: 'Where else in our environment do we see similar objects?'	DQB Question Clustering: grouping learner questions into themes shows the class how today's evidence connects to the larger phenomenon. It keeps the unresolved puzzle (area and cost) visible and motivates the next lessons.	Check that each learner has contributed at least one question and that at least some connect to measuring or testing similarity. Observe whether learners can say which questions today's work answers. Note questions that show the area and cost puzzle is still active.
Model Building Phase VIDEO:	Learners start the first draft of the class model for the passport photo. Using a sketch of S and P side by side, they label 'same	Say: 'Draw what you know so far. Show what stays the same and what changes between S and P. Then show how the distorted	Cumulative Model Building: learners draw and label a visual model that is revised each lesson. Writing what the model cannot yet	Check that models label S and P as same shape with different size and the distorted photo as not similar. Look for a sentence or

Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Scale Factor of Similar Polygons (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	shape' and 'different size'. Beside it, they sketch the distorted 6 x 4 photo and label 'stretched: not similar'. They add a note: 'Model matatu 1:25 (28 cm) is similar to the real matatu (700 cm).' They then complete the summary table: What we observed, What we learned, How it helps explain the passport photo.	photo is different.' Give WAIT TIME of 10-15 seconds before drawing starts. Circulate and ask: 'What do you still not know about how the measurements change?' Cold-call 3 learners to share one part of their model. Make clear that this is a first draft and that numbers for lengths, area and cost will be added later. Finish with an exit question: 'Name one thing around Kisumu that is similar to another and one that is not.'	explain (how length, area and cost change) points to the evidence needed in later lessons.	label that links the model matatu to similarity. Collect exit answers and check that the examples are correct.
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D. TEACHER REFLECTION

Which learners judged similarity by size rather than shape, and which pairs caused disagreement (square and rectangle, stretched flag)? Did WAIT TIME of 10-15 seconds give quieter learners time to give reasons? Were the Kenyan examples (flag, map, matatu model) familiar to all learners? How will you use today's sorting sheets to prepare learners for testing similarity with angles and side ratios in Lesson 3?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that photos S (3 x 4 cm) and P (6 x 8 cm) look the same shape but differ in size, while the distorted 6 x 4 photo looks stretched. We also saw that a flag and a smaller flag, a map and a smaller map, and a real matatu (700 cm) and its 1:25 model (28 cm) look the same shape.
What did I learn?	We learned to sort objects as similar (same shape, size may differ) or not similar (shape changed, for example stretched). A model matatu is similar to a real matatu, and 'congruent' is used only when shape and size are both the same.
How does this explain the phenomenon?	Achieng's prints S and P are similar because one is an enlargement of the other without stretching, while doubling only the width gives a distorted, non-similar photo. We still need numbers to test similarity and to explain how length, area and cost change.

LESSON 3 (40 minutes (1 double lesson of 2 x 40 minutes recommended; single-lesson version below)): Testing Similarity: Equal Angles and Proportional Sides

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners move from judging similarity by eye (Lessons 1-2) to testing it with numbers. They discover that two figures are similar only when ALL corresponding angles are equal AND all corresponding sides are in the same ratio, and they use this test to explain why Achieng's print P is the same shape as print S while the 6 x 4 cm doubled-width photo is stretched.
Knowledge	Similar figures have equal corresponding angles and proportional corresponding sides. Triangles 3-4-5 and 6-8-10 are similar (every ratio is 2). Rectangles 3 x 4 and 6 x 7 are not similar (ratios 2 and 1.75) even though all angles are 90 degrees. The distorted photo 6 x 4 cm is not similar to S (3 x 4 cm) because width ratio is 2 but height ratio is 1.
Skills	Measuring and recording corresponding sides and angles; calculating ratios (image/object) with the same units and corresponding sides in the same order; comparing ratios; using a two-part test; constructing an evidence-based argument (NGSS SEP: Analyzing and Interpreting Data; Engaging in Argument from Evidence; Using Mathematics).
Attitudes	Unity: learners work in groups and respect each other's measurements and reasoning. Responsibility: learners take care of rulers, protractors and shared photo cut-outs. Curiosity and honesty in testing a claim with evidence, even when it overturns a first impression. Appreciation of how mathematics is used in everyday photo printing.
Key Inquiry Question	1. How do we apply similarity and enlargement in day-to-day life? 2. Where else in our environment do we see objects that are similar?
Purpose in Storyline	Lessons 1-2 established that photos S and P look the same shape but differ in size, and that similar objects can be sorted by looking at shape. Lesson 3 gives learners the numerical test for similarity (equal corresponding angles, proportional corresponding sides). This sets up Lesson 4, where the common ratio is named and calculated as the linear scale factor (LSF).
Safety Notes	Use blunt-tipped geometry sets. Learners should carry scissors with closed blades if cutting card cut-outs, and keep rulers and protractors on desks when not in use. Keep aisles clear while groups move to the DQB.

B. LESSON OVERVIEW

Ms Wanjiru reminds the class of Achieng's prints in Kisumu: Mr Otieno prints S (3 x 4 cm) and P (6 x 8 cm) and they look the same shape. Learners predict how to PROVE two shapes are similar using numbers, then measure and compare triangles 3-4-5 and 6-8-10, rectangles 3 x 4 and 6 x 7, and photo S against the distorted 6 x 4 photo. They discover that equal angles alone are not enough (the rectangles), and that sides must be in the same ratio for every pair (the distorted photo fails: 2 and 1). Learners explain the two-part test, add questions to the DQB and add a first piece to the class model: a test for similarity. The name 'linear scale factor' and its use to find lengths are left for Lessons 4-5.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners look at the three pictures	Display the three photos. Say:	Prediction with commitment	Listen and check: Do learners

 VIDEO: Corresponding points and sides of scaled shapes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Corresponding Sides of Similar Triangles (PLIX) Source: CK-12  Search ARES for similar readings	on the board: photo S (3 x 4 cm), photo P (6 x 8 cm) and the distorted photo (6 x 4 cm, made by doubling only the width). Working individually for 2 minutes, then in pairs, they write a prediction in their books: 'Which pictures are the same shape as S, and what could we measure to PROVE it, not just see it?' They also predict whether two rectangles with all right angles must always be the same shape. They record a prediction on a sticky note.	'Achieng gave Mr Otieno one passport photo. He printed S and P. In Lesson 2 we sorted by eye. Today a customer says: prove it with numbers.' Ask: 'Which one looks stretched, and what exactly would you measure to show it?' Give WAIT TIME of 10-15 seconds in silence before anyone speaks. Then cold-call 4 learners from different groups (use name cards) to share a prediction without evaluating it. Ask: 'If all the corners are right angles, must two rectangles be similar? Show with your hands.' Record predictions on the board under the heading 'Our first ideas'. Do not say who is correct.	(Think-Pair-Share): writing a prediction before measuring exposes learners' prior idea that similar means looks alike, creating the cognitive conflict that the rectangles will later resolve. Learners are more invested in evidence when they have staked a claim.	suggest measuring both angles and sides? Do at least 3 of 4 cold-called learners name 'sides' or 'ratio' or 'angles'? Note learners who say 'all rectangles are similar' for revisiting in the Explain Phase. Look for predictions written on sticky notes by 100% of learners.
Observe Phase  VIDEO: Corresponding points and sides of scaled shapes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Corresponding Parts of Similar Figures (Flexbook) Source: CK-12  Search ARES for similar readings	In groups of 4 (roles: measurer, recorder, checker, reporter) learners complete 3 stations with cut-outs or drawings. Station 1: Triangles 3-4-5 and 6-8-10 cm. They measure the three angles with a protractor (about 37, 53 and 90 degrees in both) and record the side ratios in a table: 6/3, 8/4, 10/5 = 2, 2, 2. Station 2: Rectangles 3 x 4 and 6 x 7. They record all angles (90 degrees each) and ratios 6/3 = 2 and 7/4 = 1.75. Station 3: Photo S (3 x 4) and distorted photo (6 x 4). They measure angles (all 90 degrees) and compute width ratio 6/3 = 2 and height ratio 4/4 = 1. Each group completes a table with columns: Pair, Corresponding angles equal?, Ratio of side 1, Ratio of side 2, Ratio of side 3 (triangles only), Same ratios? Then they ask whether the pair looks like the same shape.	Before groups start, model one ratio: 'Image over object, same units, corresponding sides in the same order: 6 divided by 3.' Circulate and ask: 'Which side matches which? How do you know?' and 'What do you notice about your three ratios?' Remind: 'Take care of the protractors and rulers; they are shared.' Allow protractor readings within 1-2 degrees of error. Give WAIT TIME of 10-15 seconds after each probing question. Use scaffolds for learners who struggle: pre-marked corresponding sides in the same colour. For faster groups ask: 'What happens if you compare the 3 side of the first rectangle to the 7 side of the second? Is that a fair match?' Stop the stations at 18 minutes and ask each reporter to bring the table to the board.	Data table and structured comparison: organising angles and ratios in one table lets learners see patterns across cases (equal ratios plus equal angles in the triangles, equal angles but unequal ratios in the rectangles and the distorted photo). Contrasting cases build the concept of a two-part test.	Check tables for: correct pairing of corresponding sides (image over object); correct ratios 2, 2, 2 for triangles; 2 and 1.75 for rectangles; 2 and 1 for the photo; correct angle readings within 2 degrees. Observable: learners rotate or relabel shapes to match corresponding sides; learners use the same units. Note groups that match the wrong sides (for example 3 with 7) and reteach corresponding order.
Explain Phase  VIDEO: Using similarity to	Groups share findings in a gallery whole-class discussion. Learners use their tables as evidence to	Ask each reporter for 1 minute: 'What was your evidence?' Press: 'Which number tells us the shape	Claim-Evidence-Reasoning (CER) with contrasting cases: learners must back each claim with	Observable indicators: at least 80% of learners can state both parts of the test; learners cite

estimate ratio between side lengths <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Corresponding Parts of Similar Figures Practice Source: CK-12 Search ARES for similar readings	answer: 'Which pairs are similar and which are not? What is the rule?' They discuss why the rectangles 3 x 4 and 6 x 7 have all angles equal but are not similar, and why the distorted photo (6 x 4 cm) looks stretched although every angle is a right angle. They write a claim-evidence-reasoning statement: 'Triangles 3-4-5 and 6-8-10 are similar because all corresponding angles are equal and every pair of corresponding sides has ratio 2.' They then apply it to the phenomenon: 'Print P is similar to S because $6/3 = 8/4 = 2$ and angles are 90 degrees; the 6 x 4 print is not, because the ratios are 2 and 1.' Learners revisit their predictions on sticky notes and mark them 'supported', 'changed' or 'new idea'.	has been stretched?' Wait 10-15 seconds. Cold-call 3 different learners to state the rule in their own words, and then 2 more to restate it. Ask: 'Is checking angles enough? Which pair proves that?' and 'Is checking just one pair of sides enough? Which pair proves that?' (the distorted photo: width ratio is 2 but height ratio is 1). Write the two-part test on chart paper only AFTER learners state it: 1) corresponding angles equal; 2) corresponding sides in the same ratio. Say: 'Mr Otieno's customer wants a photo that is not squashed. Which test would catch a squashed print?' Be careful not to name the common ratio as the linear scale factor yet: say 'the same ratio' and tell learners that next lesson they will give it a name and use it. Do not teach area or cost here.	numbers from their tables and explain with the two-part test. Using the rectangles and distorted photo as counterexamples prevents the misconception that equal angles alone give similarity.	specific numbers (for example $7/4 = 1.75$ not 2); learners correctly say the 3 x 4 and 6 x 7 rectangles are not similar. Exit check on a slip: 'Are rectangles 2 x 3 and 4 x 6 similar? Show the ratios.' ($4/2 = 2$ and $6/3 = 2$, so yes). Identify learners confusing similar with congruent (same shape and size) and correct them.
Driving Question Board (DQB) Creation VIDEO: Side lengths after dilation <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Corresponding sides and points Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners return to the Driving Question Board with the driving question: 'How can a passport photo be made bigger or smaller without changing its shape, and how do its length, area and cost change when we do so?' In groups they write 1-2 new questions on sticky notes arising from today (for example: 'What is the name for the ratio that stays the same?', 'If sides are in ratio 2, what happens to the area?', 'Why did Mr Otieno charge KES 120 and not KES 60?', 'Where else do we see similar shapes, like maps or model matatus?'). They place notes under categories: Length, Area and cost, Everyday uses. They move answered questions (such as 'How can we tell if a photo is stretched?') to an 'Answered so far' column with evidence.	Ms Wanjiru says: 'Look at the board. What have we answered today and what new puzzles do we have?' Give 1 minute silent writing and WAIT TIME of 10-15 seconds before sharing. Cold-call 3 groups to read a question and place it, asking: 'Which part of the driving question does this help with?' Sort questions with learners: 'This is about the ratio, which Lesson 4 will name; this is about area, which we will meet later.' Mark one answered question: 'How can we tell if a photo is stretched? Test angles and sides in the same ratio.' Keep the question on how cost changes open, since it is investigated later.	Driving Question Board as a public record of growing understanding: connecting new questions to the phenomenon keeps learners motivated and shows how each lesson moves toward explaining the cost puzzle. Categorising questions helps learners see what is answered and what remains.	Count and review sticky notes: at least one question per group. Look for questions that link the ratio to area or cost (showing readiness for later lessons) versus questions that restate today's finding. Observable: learners explain placements using the words 'angles', 'sides' or 'ratio'.

Model Building Phase  VIDEO: Geometric constructions: congruent angles <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Corresponding Sides of Similar Triangles (PLIX) Source: CK-12  Search ARES for similar readings	Learners add Part 1 to the class model of how a passport photo is resized. In groups they sketch photos S and P side by side and label corresponding corners A, B, C, D and A', B', C', D' (object and image). They write beneath: 'Test for similar: 1) corresponding angles equal; 2) corresponding sides in the same ratio.' They add numbers from the evidence: 3 to 6 and 4 to 8, ratio 2 for both, all angles 90 degrees. Beside it they sketch the distorted photo 6 x 4 with a cross and write 'width ratio 2, height ratio 1: not similar'. They also write one 'still to find out' note: 'What is this ratio called, and what does it do to area and cost?'	Say: 'Our model starts with a test. Use the words object and image and label with primes: A to A'' Circulate, asking: 'Which numbers on your diagram prove the test is passed? Which prove it is failed?' WAIT TIME 10-15 seconds. Cold-call 3 groups to hold up and explain their diagrams. Create one class model on chart paper from the best elements. Say: 'This is our first draft; next lesson we will name the ratio and calculate it from S to P, S to L and P to S.' Check units (cm) and the order of corresponding sides. Close by connecting back: 'Achieng can now tell Mr Otieno exactly how to check that her postcard print is not squashed.'	Cumulative model building with drawn representations: converting measurements into a labelled diagram and test consolidates the concept and gives a visible, revisable model that later lessons extend with LSF, ASF and VSF.	Check models for: correct object-image labelling with primes; both parts of the test; ratio 2 written for S to P; the distorted photo correctly marked as not similar with ratios 2 and 1. Indicators of readiness for Lesson 4: learners write the ratio as image over object and note it is the same for all pairs of corresponding sides.
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D. TEACHER REFLECTION

Did learners understand why equal angles alone are not enough? Which groups matched corresponding sides incorrectly, and what scaffold (colour-coded sides) helped? Did learners confuse similar with congruent? How well did the 3 x 4 vs 6 x 7 rectangles create cognitive conflict? Were all learners able to compute ratios accurately (for example $7/4 = 1.75$)? Did I give 10-15 seconds of WAIT TIME consistently and cold-call across the room? What questions on the DQB show learners are ready for the scale factor in Lesson 4?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We measured the angles and compared sides of triangles 3-4-5 and 6-8-10 (all ratios 2), rectangles 3 x 4 and 6 x 7 (ratios 2 and 1.75), and photo S (3 x 4 cm) against the distorted 6 x 4 cm photo (ratios 2 and 1). All the rectangles and photos had right angles.
What did I learn?	Figures are similar only if corresponding angles are equal AND corresponding sides are in the same ratio. Equal angles alone are not enough, and one matching ratio is not enough.
How does this explain the phenomenon?	Print P (6 x 8 cm) is similar to print S (3 x 4 cm) because all angles are 90 degrees and $6/3 = 8/4 = 2$. The photo made 6 x 4 cm is stretched because the width ratio is 2 but the height ratio is 1, so it is not similar. This helps explain how Achieng's photo can change size without changing shape.

LESSON 4 (40 minutes): Finding the Linear Scale Factor: How Many Times Bigger Is Achieng's Photo?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners calculate the linear scale factor $k = \text{image length} / \text{object length}$ from corresponding lengths of Achieng's passport photo prints (S to P, S to L, P to S) and write the first draft of an enlargement model that explains how lengths change when the photo is resized.
Knowledge	LSF $k = \text{image length} / \text{object length}$, using the same units and corresponding sides in the same order. S to P: $k = 2$ ($6/3 = 8/4 = 10/5 = 4.8/2.4 = 2$). S to L: $k = 3$ ($9/3 = 12/4 = 3$). P to S: $k = 1/2$ ($3/6 = 4/8 = 1/2$). $k > 1$ is an enlargement and $0 < k < 1$ is a reduction. The image is labelled with a prime (A to A').
Skills	Calculating ratios of corresponding lengths; checking that all pairs of corresponding sides give the same k ; developing and using a model (NGSS SEP 2); analysing and interpreting data (NGSS SEP 4); using mathematics and computational thinking (NGSS SEP 5).
Attitudes	Accuracy and care in measuring and calculating; collaboration in group discussion; curiosity about how everyday objects such as passport photos follow mathematical rules; responsibility when handling rulers and shared print cards.
Key Inquiry Question	How do we apply similarity and enlargement in day-to-day life?
Purpose in Storyline	Lesson 3 showed that similar figures have equal corresponding angles and corresponding sides in the same ratio. Lesson 4 names that constant ratio as the linear scale factor and gives learners a number (k) to describe how Mr Otieno's prints S, P and L relate. It also launches the first draft of the model, so Lessons 5 to 7 can extend it to unknown lengths, area (and cost) and volume. The model is deliberately incomplete: it explains lengths but not yet why the cost rises from KES 30 to KES 120.
Safety Notes	Learners use rulers and paper print cards only. Keep walkways clear while groups move to the gallery wall. Rulers are used for measuring only, not for play. Sharpened pencils stay on the desks.

B. LESSON OVERVIEW

Achieng's passport photo is printed by Mr Otieno in Kisumu as a stamp print S (3 cm x 4 cm) and a postcard print P (6 cm x 8 cm). In Lesson 3 learners tested similarity and found that corresponding sides are in the same ratio. Today they name this ratio the linear scale factor (LSF). Learners predict how many times bigger P is than S, measure and calculate $k = \text{image/object}$ for width, height, diagonal and face height, then extend the idea to S to L ($k = 3$) and reverse it to P to S ($k = 1/2$). They use the distorted 6 x 4 photo as a contrast case: the width ratio is 2 but the height ratio is 1, so there is no single k . Finally learners write the first draft of the enlargement model (a labelled diagram plus a rule in words and symbols) and add questions to the Driving Question Board (DQB) about area and cost, which the model cannot yet explain.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Dilations: scale factor Source: Khan	Learners recall the DQB question: How can a passport photo be made bigger or smaller without changing its shape, and how do its length, area and cost change	Display the DQB and say: "Yesterday we showed that corresponding sides of similar figures are in the same ratio. Today we put a name and a	Prediction with competing ideas (Predict-Observe-Explain). Writing a prediction and reason before seeing the data makes learners commit to a prior idea (additive	Observe whether learners write a prediction with a reason, not just a number. Look for the additive misconception (e.g. 'S + 3 cm = P') and tally on a clipboard how many

Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar readings	when we do so? Each learner receives a card showing Achieng's prints S (3 x 4 cm) and P (6 x 8 cm) with the width of S and P marked. Individually (2 minutes) they answer: 'P looks bigger than S. By what number could we describe how many times bigger it is? Will the same number work for the height and the diagonal too?' They write a prediction and a reason on a sticky note, then do a pair share. Some learners say 'add 3', others say 'times 2'. They also predict: 'If Achieng asked Mr Otieno to go from P back to S, what would the number be?'	number to that ratio." Hand out the cards. Say: "Mr Otieno printed the same photo twice. Without calculating yet, what single number tells us how many times bigger P is than S? Write it down with a reason." WAIT TIME: allow 10-15 seconds of silent thinking before writing begins. Circulate and note the different ideas (adding versus multiplying). Cold-call 3 learners, each from a different group, to share a prediction without saying whether it is right. Record 'add 3' and 'times 2' on the board as competing ideas. Then ask: "What would we need to measure to test these ideas?"	versus multiplicative thinking) that the evidence can confront. The conflict between 'add' and 'times' creates a need to know and a reason to calculate carefully.	use adding versus multiplying. Indicator of readiness: at least a third of learners mention 'times' or 'ratio' linked to Lesson 3 before the Observe Phase. Note learners who say the number will be different for the height and the diagonal, to be revisited in the Observe Phase.
Observe Phase VIDEO: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar readings	In groups of 4, learners get print cards S, P and L (each a labelled rectangle with a diagonal and face height) and a ruler. Step 1: they measure the width, height, diagonal and face height of S and P, and complete a table: Length S (object) P (image) P/S. They find $6/3 = 8/4 = 10/5 = 4.8/2.4 = 2$ and name $k = 2$ as the linear scale factor for S to P. Step 2: they repeat for S to L ($9/3 = 12/4 = 3$, $k = 3$). Step 3: they reverse the direction, P to S (object P, image S): $3/6 = 4/8 = 1/2$ and note that $k < 1$ is a reduction. Step 4: contrast case. They measure the distorted photo (6 cm x 4 cm) and calculate width $6/3 = 2$ and height $4/4 = 1$, so no single k exists and the picture looks stretched. They also check that angles at the corners stay 90° in all prints. Each group writes the rule: $k = \text{image length} / \text{object length}$.	Model once with the width of S and P on the board: "Image length goes on top, object length on the bottom. Which print is the object here? Which is the image?" Circulate with the prompts: "Which sides are corresponding? How do you know?" and "Is your unit the same on top and bottom?" After 5 minutes, say: "Stop and look at your last column. What do you notice?" WAIT TIME: 10-15 seconds of silent scanning. Cold-call 4 groups, one for each row (width, height, diagonal, face height), to report their ratio. Then ask: "Reverse the direction, from P to S. What happens to the number? Why is it a fraction?" Introduce the prime notation: "If the corner of S is A, the matching corner of P is A'." For the distorted photo ask: "Can we find one k that works for both width and height? What does that tell us about the shape?" Reinforce: the order of the two lengths matters, because k depends on which print is the	Structured data table and pattern-seeking (analysing and interpreting data). Putting every ratio in one column lets learners see that all pairs give the same constant, which is exactly what makes k a single number. The reversed direction (P to S) and the distorted photo are contrasting cases that sharpen the meaning: the direction fixes k, and a single constant k is what keeps the shape unchanged.	Check each group's table: all four ratios for S to P equal 2; for S to L equal 3 (width and height) and for P to S equal 1/2. Indicators: learners put image over object, not object over image; they keep the same units; they identify corresponding sides in order. Ask 2 random learners to point to the object and image in their table. Common errors to note: dividing S by P and getting 1/2 for S to P; measuring the face height inaccurately; subtracting lengths. Listen for the statement 'the distorted photo has two different ratios, so it is not an enlargement of S'.

		object.		
Explain Phase  VIDEO: Using similarity to estimate ratio between side lengths <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: center <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners work in pairs on a set of short tasks and then explain in a class discussion. Task 1: Achieng's friend asks Mr Otieno for a print that is 3 times S. Write k, and say how the width and the height change (3 to 9 and 4 to 12). Task 2: Mr Otieno says 'P is an enlargement of S with $k = 2$.' Achieng's classmate says 'the face got 2.4 cm bigger.' Learners use the numbers (2.4 to 4.8) to explain who is right and why (it is 2 times, not +2.4). Task 3: Sort three descriptions as enlargement or reduction: $k = 3$, $k = 1/2$, $k = 2$. Task 4: Using any corresponding pair, write the sentence: 'Image length = $k \times$ object length' and test it on the diagonal of S and L (5 cm to 15 cm). Learners write one explanation for the question: Why does the shape stay the same when every length is multiplied by the same k?	Say: "Use your table as evidence. I want a number and a reason in each answer." During Task 2 ask: "Is the gain in face height the same as the gain in width? So can adding be the rule?" Press for reasoning: "How do you know the ratio works for every length, not just the width?" Use WAIT TIME of 10-15 seconds after every question before taking answers. Cold-call 3 pairs to give their explanation for Task 4, and ask a fourth learner to restate it in their own words. Link back to the Predict Phase: "Whose prediction, 'add 3' or 'times 2', is supported by the evidence? Which data point shows it?" Address the misconception with a counter example: adding 3 to the width would give 6, but adding 3 to the height gives 7, not 8. Close by writing on the board: $k = \text{image length} / \text{object length}$, $k > 1$ enlargement, $0 < k < 1$ reduction.	Claim-Evidence-Reasoning with revisiting of predictions. Learners state a claim (P is an enlargement of S with $k = 2$), cite measured evidence (all four ratios), and reason with the idea of a constant multiplier. Comparing additive and multiplicative explanations explicitly repairs the misconception and makes learners use Lesson 3's proportional sides to justify the new idea.	Collect Task 2 and Task 4 explanations. Success indicators: the response cites at least two corresponding lengths with their ratio, states that every length is multiplied by the same k, and rejects an additive rule using numbers. Quick check by show of fingers: 'What is k from S to L?' (3 fingers) and 'What is k from P to S?' (a learner says 1/2). Note learners who still reverse object and image and give them one more example from the Kisumu shop.
Driving Question Board (DQB) Creation  VIDEO: Dilations: center <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations and properties <i>Related topic: not an exact match</i>	Learners return to the DQB led by Ms Wanjiru. They write on cards what they now know (e.g. 'S to P has $k = 2$, so every length doubles', 'P to S has $k = 1/2$') and new questions that arose. Each group posts at least one new question, for example: 'The width and height doubled, but why did the price go from KES 30 to KES 120, which is more than double?', 'Does the area also double when $k = 2$?', 'If the large print L has $k = 3$, how much will Mr Otieno charge?', and 'If we know k, can we find a missing length?' Learners put their cards under headings: Lengths (answered), Area and cost (not yet answered), Unknown lengths,	Say: "Look at the DQB. Which of our questions can we answer now, and which must wait?" Give each group 3 minutes to write. WAIT TIME: 10-15 seconds after asking, "What is the puzzle that LSF alone cannot solve?" Cold-call 3 groups to read out a question and say where it belongs on the board. Highlight the cost question: "Mr Otieno charges KES 2.50 per cm^2 of paper. Lengths doubled, but KES 30 became KES 120. LSF explains lengths, so what is missing?" Deliberately do not answer; tell learners this is for later lessons. Move answered questions to the 'Answered' column with a learner's help and	Driving Question Board with a need-to-know sort. Sorting questions into answered and open ones shows learners the growth of their understanding and points at the gap: lengths scale by k, but cost does not scale by k. This keeps the phenomenon at the centre and motivates the investigation of area in the coming lessons.	Review the cards posted. Indicators: at least one question per group links to area or cost; questions are phrased with numbers from the anchor data (e.g. 30, 120, 2.50); learners place cards under the correct heading. Watch for learners who ask only 'what is k?' type questions, who may need a further example. Record 2 or 3 strong questions to use as launch points for Lesson 6.

for this lesson. Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Volume. They place a green dot on answered questions, and a yellow dot on those still open.	tag the new question about unknown lengths for Lesson 5.		
Model Building Phase  VIDEO: Dilating shapes: shrinking <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Dilate triangles <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Individually, then in pairs, learners draw the first draft of their enlargement model on a half-page of paper: (1) a diagram of S (3 x 4 cm) and P (6 x 8 cm) with corresponding corners labelled A, B, C, D and A', B', C', D' and arrows marked 'x 2'; (2) a rule in words and symbols: $k = \text{image length} / \text{object length}$, image length = $k \times \text{object length}$; (3) notes on $k > 1$ (enlargement) and $0 < k < 1$ (reduction), with P to S as $k = 1/2$; (4) a box 'What my model cannot explain yet' where they write the cost puzzle: 'KES 30 to KES 120 is not x 2'. Pairs swap drafts and add one comment for improvement. Learners keep the draft in their folder for revision in later lessons.	Say: "A model must help us explain and predict. Your first draft should predict the width and height of print L from S." Ask pairs to test the model: "Use $k = 3$. What are the width and height? Does your model give 9 and 12?" WAIT TIME: 10-15 seconds before pairs begin the swap. Circulate and ask: "Where in your model could the price appear? What would you need to add?" Cold-call 3 learners to read out the box on what the model cannot yet do. Provide the sentence stem: "My model shows _____ but it cannot yet explain _____ because _____." Collect the drafts as exit slips or keep them in folders. Remind learners that Lesson 5 will use the model to find unknown lengths.	Developing and using models (NGSS SEP 2) through an explicit first draft with limitations. Learners represent the relationship in a diagram and a symbolic rule, test it by predicting L, and identify its gaps. Naming what the model cannot explain creates a purpose for iterative revision as area and volume are added.	Check each draft against a four-point checklist: (a) both prints are drawn with correct dimensions and labelled corners using primes; (b) rule $k = \text{image/object}$ is correct; (c) enlargement and reduction are distinguished using P to S = $1/2$; (d) the limitation about cost is stated. Indicator of mastery: the model correctly predicts L as 9 x 12 using $k = 3$. Learners who write $k = \text{object/image}$ or omit the units get one-to-one feedback.

D. TEACHER REFLECTION

After the lesson, reflect on: (1) How many learners moved from an additive idea ('add 3') to a multiplicative one ('times 2') and what evidence in the tables showed this? (2) Did learners keep image over object consistently, especially in the P to S reduction? (3) Did the distorted 6 x 4 photo help learners see that a single k is what preserves the shape? (4) Which DQB questions about area and cost were strongest, and how will you use them to launch Lesson 6? (5) Which learners need support with measuring accurately or with fractions such as $1/2$? Plan a short follow-up with extra Kisumu examples for those learners before Lesson 5.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We measured width, height, diagonal and face height of Achieng's prints S (3 x 4 cm), P (6 x 8 cm) and L (9 x 12 cm). For S to P, every length was 2 times bigger ($6/3 = 8/4 = 10/5 = 4.8/2.4 = 2$), and for S to L it was 3 times. In the distorted 6 x 4 photo, the width ratio was 2 but the height ratio was 1.
What did I learn?	The linear scale factor $k = \text{image length} / \text{object length}$, using the same units and corresponding sides in the same order. $k > 1$ is

	an enlargement and $0 < k < 1$ is a reduction (P to S has $k = 1/2$). A single constant k for all lengths keeps the shape the same.
How does this explain the phenomenon?	Print P is an enlargement of print S with LSF 2, so every length doubles by multiplying, not by adding, and the shape stays the same. Our first model explains lengths but not yet why Mr Otieno's price went from KES 30 to KES 120, so we need to investigate area next.

LESSON 5 (40 minutes): Using LSF to Find Unknown Lengths: Photo L and the Kericho Tree Shadow

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use the linear scale factor (LSF) from Lesson 4 to calculate missing sides in similar figures, including photo L from photo S and the height of a tree in Kericho from shadows. This moves the Driving Question forward: if we know the LSF, we can predict every unknown length in a resized passport photo.
Knowledge	If two figures are similar, every missing length on the image = object length $\times k$, and every missing length on the object = image length $\div k$. When no k is given, it is found from one pair of corresponding sides whose lengths are both known ($k = \text{image/object}$, same units, corresponding sides in the same order).
Skills	Identify corresponding sides; calculate k from a known pair; use k to find an unknown side by multiplying or dividing; set up similar triangles from shadows; check answers for reasonableness (Developing and Using Models; Using Mathematics and Computational Thinking; Constructing Explanations).
Attitudes	Appreciate the usefulness of similarity in measuring things we cannot reach, such as a tall tree; show persistence and accuracy when working with units; cooperate respectfully in pairs and groups.
Key Inquiry Question	How do we apply similarity and enlargement in day-to-day life? Where else in our environment do we see objects that are similar?
Purpose in Storyline	Lesson 4 taught learners to calculate $\text{LSF} = \text{image/object}$ from corresponding lengths (S to P $k = 2$, S to L $k = 3$, P to S $k = 1/2$). Lesson 5 turns LSF into a tool to find missing lengths: photo L from S, and a tree from its shadow. This prepares Lesson 6, where learners investigate how area changes (ASF) and why the cost puzzle in Mr Otieno's shop arises.
Safety Notes	Shadow measurement is done in the school compound or from a prepared diagram. Learners stay on the school grounds, do not climb trees or stand on desks, and avoid looking directly at the sun. Metre rules and tape measures are handled carefully and not swung.

B. LESSON OVERVIEW

Achieng in Kisumu has photo S (3 cm x 4 cm). Mr Otieno offers a large print L but the order slip has only the LSF. Can Achieng find the width and height of L before paying? Learners predict, then use LSF to find missing sides of photo L (9 cm x 12 cm) from S. They then move to a Kericho scene where a 1.5 m student casts a 2 m shadow while a tree casts a 12 m shadow. Learners find $k = 6$ and the tree height of 9 m. In the Explain phase they build a general method (identify corresponding sides, find k , multiply or divide). Learners then add to the Driving Question Board (DQB) and revise the draft enlargement model from Lesson 4 so it can find unknown lengths.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Dilations: scale factor Source: Khan	Learners are shown Achieng's problem: photo S is 3 cm wide x 4 cm tall. Mr Otieno says print L is an enlargement of S with LSF 3. Individually in notebooks, learners	Display the S print dimensions and say: 'Achieng has photo S, 3 by 4 centimetres. Mr Otieno will print L with LSF 3. Before we calculate, write your prediction for the width	Prediction with reasoning (Predict-Observe-Explain): writing a prediction and the reason for it makes current ideas visible, including the common	Look for: (1) learners who multiply 3×3 and 4×3 versus those who add 3; (2) sketches showing the student and tree as similar triangles or as unrelated lines; (3)

Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar readings	predict the width and height of L and explain in one sentence how they got them. Then they predict: 'In Kericho, a 1.5 m student casts a 2 m shadow. A tree casts a 12 m shadow at the same time. Is the tree taller or shorter than 12 m? By how much, roughly?' Learners draw a quick sketch of the student, tree and shadows.	and height of L. Do not talk yet.' Give 2 minutes of silent writing. Then ask: 'Who thinks the tree is taller than its shadow? Who thinks it is shorter? Show with a thumbs up or down.' WAIT TIME 10-15 seconds after each question before taking responses. Cold-call 3 learners to share predictions, asking each: 'What made you choose that number?' Record the range of predictions on the board without judging. Note learners who add 3 to each side (additive thinking) instead of multiplying.	misconception of adding instead of multiplying. Learners return to these predictions at the end to see how their thinking changed.	whether learners link the prediction to the LSF idea from Lesson 4. Tally on the board how many predicted 9 x 12 and how many predicted other sizes.
Observe Phase VIDEO: Scale drawings Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Scale drawings Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Activity A (photo L, 12 min): In pairs, learners receive a table with columns Object (S) and Image (L). They use LSF 3 to fill in width (3 to ?), height (4 to ?), and diagonal (5 to ?). They also reverse the question: 'Mr Otieno's half-size print H has a height of 2 cm. Using S to H, what is its width?' ($k = 1/2$, width 1.5 cm). They then check that $9/3 = 12/4 = 15/5 = 3$. Activity B (shadows, 12 min): In groups of four, learners work from the Kericho diagram (or, if possible, measure a pole and its shadow outside). Data: student 1.5 m, student's shadow 2 m, tree shadow 12 m. They label the two triangles, match corresponding sides (shadow to shadow, height to height), calculate $k = 12/2 = 6$ and then find the tree height = $1.5 \times 6 = 9$ m. Groups record what they did at each step.	Circulate with a checklist. Say: 'Before you multiply, point to the object side and the matching image side. Which side matches which?' For groups stuck on shadows, ask: 'Which two lengths are both shadows? What is the scale factor from the small shadow to the big shadow?' Do not give the answer. WAIT TIME 10-15 seconds after each question before giving a hint. Ask 2 groups to explain why we use shadow-to-shadow to find k when the height of the tree is unknown. Remind learners: 'Keep the same units, and keep the same order: image over object.' Stop the class after Activity A and cold-call 2 pairs to report the width and height of L before moving to Activity B.	Structured data table and diagram annotation: organising corresponding lengths in a table and labelling the diagram helps learners see that the same multiplier works for every side. Comparing the photo and shadow problems shows that the same structure appears in two different contexts.	Observable indicators: learners correctly write L as 9 cm x 12 cm with diagonal 15 cm; learners correctly give H's width as 1.5 cm by using $k = 1/2$; learners match shadow to shadow and height to height; learners write $k = 12/2 = 6$ and the tree height as 9 m with units. Note any group dividing 1.5 by 6 or mixing up object and image.
Explain Phase VIDEO: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES	In their groups, learners write a 'method card' with the steps for finding an unknown length in similar figures. They then use the method on a new problem: 'A pole in Kisumu casts a 3 m shadow while a nearby 1.5 m student casts	Say: 'Write a method card that another Grade 10 class could use without you. What goes first? What goes second?' Take 3 cold-calls to read out steps, and build a class version on the board: (1) Find a pair of corresponding sides with	Generating a method card (explanation in learners' own words) and a transfer problem: writing the procedure forces learners to state the general rule, while the new pole problem checks that they can apply it in a fresh	Check method cards for: use of corresponding sides, image/object order, same units, and the multiply/divide distinction. Check the pole answer of 4.5 m. A learner who writes $k = 1/3$ or gets 0.5 m is mixing up object and

for similar videos EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar readings	a 1 m shadow. How tall is the pole?' ($k = 3/1 = 3$, pole = $1.5 \times 3 = 4.5$ m). They then explain why the tree could not simply be 12 m tall like its shadow, using the idea of equal angles of the sun's rays and the same shape of the two triangles. Each group links the method back to the Driving Question by writing one sentence: 'To find any missing length in a resized passport photo, we ...'	both lengths known. (2) Calculate $k = \text{image/object}$. (3) Multiply object lengths by k to get image lengths, or divide image lengths by k to get object lengths. (4) Check with another pair of sides. Ask: 'When do we multiply and when do we divide?' WAIT TIME 10-15 seconds, then cold-call 2 learners. Ask: 'Why does the sun make the student triangle and the tree triangle similar?' Accept answers about equal angles and the same time of day. Press for evidence: 'What do you see in the diagram that shows that?'	context. Claim-Evidence-Reasoning is used when explaining why the triangles are similar.	image. Collect 2 or 3 cards as exit evidence of understanding.
Driving Question Board (DQB) Creation VIDEO: Scale drawings Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Scale drawings Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners add new sticky notes to the DQB under two headings: 'What we figured out' and 'New questions'. Examples of what they have figured out: 'If we know LSF, we can find any missing length of the photo,' 'A tree's height can be found from shadows using LSF 6.' Likely new questions: 'The lengths scale by 3, but does the area scale by 3 too?' 'Why does Mr Otieno charge KES 120 and not KES 60 for P?' 'Does this work for real maps?' Learners place questions near related earlier questions.	Say: 'Ms Wanjiru's board is our record of thinking. Write one thing we now know and one thing we still wonder. One idea per note.' Give 3 minutes. Invite 4 learners to post and read their notes. Ask: 'Which of our earlier questions has this lesson helped us answer?' WAIT TIME 10-15 seconds. Circle the area and cost questions and say: 'We now know lengths follow LSF. Next lesson we will test what happens to area and cost.' Do not answer the area question yet.	Driving Question Board as a public record: linking new questions to previous ones shows learners how today's evidence answers some questions and creates others, keeping the phenomenon in view.	Observe whether notes are linked to evidence (the 9 m tree, the 9 x 12 photo) and whether at least some learners raise the area/cost question without prompting. Count how many questions go beyond length (area, cost, volume).
Model Building Phase VIDEO: Using similarity to estimate ratio between side lengths <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English	Learners revise the first draft of the enlargement model from Lesson 4. Individually, each learner adds a 'Finding unknown lengths' box with a diagram of S and L (3×4 to 9×12, $k = 3$) and a diagram of the student and the tree ($k = 6$, tree 9 m), plus the rule: image = object $\times k$, object = image / k. Learners add a 'Still to explain' box with a question mark for area and cost, using the numbers they already	Say: 'Open your model from Lesson 4. Add what today's evidence lets you predict. Include one diagram and one rule.' Circulate and ask: 'If Mr Otieno gave you only the width of P, how would you find its height?' ($k = 6/3 = 2$, height $4 \times 2 = 8$ cm). WAIT TIME 10-15 seconds. Cold-call 3 learners to read their rule aloud. Close by asking: 'Lengths doubled from S to P. Does the cost, KES	Cumulative model revision: adding to the same model each lesson shows learners how the evidence accumulates, and the 'Still to explain' box highlights the gap that Lesson 6 will address.	Check the model for: the correct rule (image = object $\times k$; object = image / k), both worked examples with correct values (9×12 cm; 9 m tree), and a clear note that area and cost are not yet explained. Learners who can find the height of P from its width only demonstrate the transfer needed.

- US curriculum) Search ARES for similar videos EXERCISE: Prove triangle similarity <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	know: P costs KES 120 and S costs KES 30, but the LSF is 2. Finally, learners return to their predictions and note one thing they got right and one they would change.	30 to KES 120, just double? What do we need to find out?' Leave the question open for Lesson 6.		
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D. TEACHER REFLECTION
Did learners correctly identify corresponding sides before calculating, especially in the shadow diagram? How many still added instead of multiplying, and what evidence helped them change? Which learners could explain why the sun's rays create similar triangles? Did the cost question emerge from learners themselves? Note groups who need more practice with image/object order and unit consistency before Lesson 6.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Photo L is 9 cm x 12 cm with diagonal 15 cm when S (3 x 4, diagonal 5) is enlarged with LSF 3. A 1.5 m student with a 2 m shadow and a tree with a 12 m shadow give $k = 12/2 = 6$.
What did I learn?	If figures are similar, we find k from one pair of corresponding sides (image/object), then multiply object lengths by k to get image lengths, or divide image lengths by k to get object lengths. The tree height is $1.5 \times 6 = 9$ m.
How does this explain the phenomenon?	Shadows work because the sun's rays make equal angles, so the student and tree triangles are similar. Every length in the photo follows the same LSF, so we can predict any missing side of a resized passport photo. We have not yet explained why the cost of P is KES 120 rather than double KES 30.

LESSON 6 (40 minutes (one lesson; double lesson of 80 minutes if grid-paper investigation is extended)): Area Scale Factor and the Cost Puzzle: Why Does Doubling the Photo Quadruple the Price?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate on grid paper how the area of an enlarged figure relates to the linear scale factor, establish that $ASF = k^2$, and use it to explain why Mr Otieno's charge for print P is four times, not double, the charge for print S.
Knowledge	By the end of the lesson the learner should be able to: (a) state that when a figure is enlarged with linear scale factor k , its area is multiplied by the area scale factor $ASF = k^2$; (b) calculate ASF from LSF and LSF from ASF for the photo prints S, P, L and H; (c) explain that paper cost follows area (KES 2.50 per cm^2), not length.
Skills	Predicting, counting unit squares on grid paper, calculating area ratios, identifying patterns in a table, constructing an evidence-based explanation (NGSS SEP: Analyzing and Interpreting Data; Using Mathematics and Computational Thinking; Constructing Explanations).
Attitudes	Communication and Collaboration: learners discuss and justify predictions with peers. Critical Thinking and Problem Solving: learners test their own predictions against evidence. Responsibility: learners take care of shared grid paper and rulers. Unity: learners work in groups and share findings. Life Skills (self-esteem): learners participate confidently and accept that revising a wrong prediction is part of learning.
Key Inquiry Question	How do we apply similarity and enlargement in day-to-day life? (Here: pricing printed photos by area.) Where else in our environment do we see objects that are similar? (Photos, posters and signboards that are enlarged copies.)
Purpose in Storyline	Lessons 4-5 gave learners the linear scale factor and the use of LSF to find lengths. In Lesson 1 learners predicted how cost would change when photo S became photo P. This lesson answers the cost part of the driving question: lengths double (LSF 2) but area is multiplied by 4, so the price goes from KES 30 to KES 120. It sets up Lesson 7, where learners test whether the pattern continues for volume (k^3).
Safety Notes	Low risk. Scissors, if used to cut grid paper, are used at desks only. Keep bags and walkways clear during group work. Rulers are used for measuring only.

B. LESSON OVERVIEW

Achieng notices that Mr Otieno's print P (6 cm x 8 cm) has sides twice as long as print S (3 cm x 4 cm), but it costs KES 120, not KES 60. Learners first revisit their Lesson 1 cost predictions and then predict how many copies of print S would fit on print P. On centimetre grid paper they draw the outlines of S (3 x 4), P (6 x 8), L (9 x 12) and H (1.5 x 2), count squares and fill a table of LSF, area and ASF. They see $ASF = k^2$ (1, 4, 9, 1/4 for $k = 1, 2, 3, 1/2$). They then connect area to price at KES 2.50 per cm^2 (30, 120, 270, 7.50) and explain why cost is more than doubled. They test the idea on the distorted 6 x 4 cm photo (area 24 cm^2) to see that the rule k^2 holds only when both dimensions are scaled by the same k . The DQB is updated and the class adds an ASF rule to the draft enlargement model started in Lesson 4.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	(8 min) Ms Wanjiru displays the	Hold up the two prints and say:	Predict-Observe-Explain (the P	Look for: (1) the number of

 VIDEO: Scale factors and area Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Construct scale drawings Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Kisumu scenario: Achieng's print S (3 x 4 cm, KES 30) and print P (6 x 8 cm, KES 120). Learners recall that the LSF from S to P is 2 and that every length doubles. Each learner silently writes answers on a prediction card: (1) How many copies of print S do you think would fit exactly on print P? Draw your guess. (2) If the lengths double, by what number is the area multiplied? (3) Why does Mr Otieno charge KES 120 and not KES 60? Learners then compare their Lesson 1 cost prediction with the new one in pairs and put a tick or a question mark next to each guess. Learners also predict the area scale factor when LSF is 3 (print L, 9 x 12 cm).	"Achieng's sides doubled, yet the price did not just double. Before we investigate, write what you think is happening." Give 2 minutes of silent writing, then WAIT TIME of 10-15 seconds before any talk. Say: "Turn to your partner and compare. Where do you disagree?" Cold-call 4 learners with different answers (for example 'area doubles', 'area is 4 times', 'not sure'), and record every prediction on the board without saying which is right. Ask: "What would you need to measure to test this?" Ask 2 learners to refer to their Lesson 1 cost prediction and say whether they would change it.	step) with Public Prediction Recording: writing predictions before data makes learners commit to ideas and exposes the common misconception that area doubles when lengths double. Learners later compare evidence against their own wording, which builds ownership of the revised idea.	learners who predict area x2 versus x4 (tally on the board); (2) whether learners draw copies of S on P with sensible placement; (3) whether prediction cards give a reason connected to LSF or just a guess. Learners who say 'the price doubles because the sides double' are flagged for targeted questions in the Observe phase.
Observe Phase  VIDEO: Scale factors and area Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Construct scale drawings Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(12 min) In groups of 4 with centimetre grid paper, a ruler and pencil, learners (1) draw rectangle S (3 cm x 4 cm) and count its squares to get 12 cm²; (2) draw P (6 x 8) and count or calculate squares to get 48 cm²; (3) test their prediction by cutting or drawing copies of S inside P and finding that exactly 4 copies fit; (4) draw L (9 x 12) and find 108 cm², noting that 9 copies of S fit; (5) draw H (1.5 x 2) using half squares, find 3 cm², and note that 4 copies of H fit inside S. They record in a table: print, LSF from S, length of base, area, ASF = area of image / area of S. The data are: S k=1 area 12 ASF 1; P k=2 area 48 ASF 4; L k=3 area 108 ASF 9; H k=1/2 area 3 ASF 1/4. Groups then add a column k² and compare it to ASF. Finally they add the distorted photo (6 x 4 cm, area 24 cm², only width doubled): the area is only 2 times, and ask why it does not match.	Circulate and say: "Count the squares in a way that you can explain to another group. Is there a faster way than counting one by one?" Ask groups stuck on H: "What does 1.5 cm look like on the grid? How can half squares be combined?" Cold-call 3 groups to report the area of S, P and L, and write 12, 48, 108 and 3 in the class table. After the table is filled give WAIT TIME of 10-15 seconds and ask: "What do you notice when you compare the ASF column with the LSF column?" Do not give the rule yet. For the distorted photo ask: "Both width and height must change by the same factor for this rule. What happened to the height here?" Prompt: "Where in Kisumu have you seen a stretched photo or poster that looks wrong?"	Structured data table with Pattern Noticing: learners build the table themselves from counted evidence, so the relationship ASF = k² emerges from their own data rather than being told. Comparing the distorted photo as a contrasting case shows the boundary of the rule.	Check that each group has correct areas 12, 48, 108 and 3 cm² and correct ASF values 1, 4, 9 and 1/4. Observable indicators: learners count 4 copies of S inside P without gaps; learners write the ASF as a number, not a length; learners notice ASF equals k times k. Note groups who still record ASF as 2 for P and re-ask: "How many copies of S did you actually count?"

Explain Phase  VIDEO: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(10 min) Each group writes a claim-evidence-reasoning statement: Claim: when LSF is k, ASF is k^2. Evidence: data from the table ($k=2$ gives 4; $k=3$ gives 9; $k=1/2$ gives $1/4$). Reasoning: when both length and width are multiplied by k, area = $(k \times \text{length}) \times (k \times \text{width}) = k^2 \times \text{original area}$. Learners then use the pricing rule (KES 2.50 per cm^2) to calculate costs: S $12 \times 2.5 = \text{KES } 30$; P $48 \times 2.5 = \text{KES } 120$; L $108 \times 2.5 = \text{KES } 270$; H $3 \times 2.5 = \text{KES } 7.50$. They check the cost ratios: $120/30 = 4 = 2^2$ and $270/30 = 9 = 3^2$. They write a two-sentence explanation of Achieng's puzzle: why KES 120 and not KES 60. They also explain why the distorted photo (24 cm^2, KES 60) is only twice the area and is not an enlargement of S. Finally they answer: if Achieng wanted a print with LSF 3, how much more than S would she pay? (9 times, KES 270.)	Say: "Use your own table to complete this sentence: when the lengths are multiplied by k, the area is multiplied by ____." After WAIT TIME of 10-15 seconds, cold-call 3 learners to give the sentence, then ask a fourth to say it in their own words. Press for reasoning: "Why k times k and not k plus k?" Draw a 3×4 rectangle and a 6×8 rectangle on the board, labelling both dimensions doubled. Ask: "Mr Otieno charges KES 2.50 per cm^2. Does he charge by length or by area? How do you know?" Address the misconception: "If a learner says the cost should be KES 60 because the photo is twice as big, what would you show them?" Make sure learners use 'ASF' and 'LSF' correctly and that 'similar' is used for same shape.	Claim-Evidence-Reasoning (CER) with Revising Predictions: learners tie the claim to their own measured evidence and then reconcile it with their earlier predictions, shifting from 'area doubles' to 'area is multiplied by k^2'. Applying it to cost links the abstract rule to the phenomenon.	Indicators of understanding: learners state $\text{ASF} = k^2$ with evidence; learners compute ASF from k in a new case (for example $k = 5$ gives 25) and compute k from ASF (ASF 9 gives $k = 3$); learners explain cost by area rather than length. Quick exit check on a slip: 'Print L has LSF 3 from S. What is its ASF and cost?' (Answers: 9 and KES 270.) Learners who cannot answer receive a re-teach with the grid-paper drawing.
Driving Question Board (DQB) Creation  VIDEO: Dilating triangles: find the error <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Similarity & transformations <i>Related topic: not an exact match for this lesson.</i>	(5 min) Learners return to the Driving Question Board begun in Lesson 1. Using sticky notes, they (a) place answers or evidence under the question 'Why does the photo cost more than double?' with the note '$\text{ASF} = k^2$, so cost $\times 4$ for $k=2$'; (b) move any answered questions to an 'Answered' column; (c) write new questions that arise, such as: What happens to the volume of a jerrycan if its sides double? What if the LSF is $1/2$: does the price fall to a quarter? How do we draw an enlargement if we have a centre?	Say: "Which of our DQB questions can we now answer with evidence from today? Come and place your note." Cold-call 3 learners to read a sticky note aloud and justify its place. WAIT TIME of 10-15 seconds before asking: "What new question has today created that we cannot yet answer?" Highlight the question about volume and promise: "We will test it in the next lesson with cubes and jerrycans." Link back to Ms Wanjiru's first DQB entries from Lesson 1.	Driving Question Board as a living record of progress: moving questions between 'unanswered' and 'answered' with evidence helps learners see the storyline of the unit and fuels motivation for Lesson 7.	Check that sticky notes cite evidence (areas, ASF, cost) and not just opinions. Note the number of new, relevant questions (volume, fractional k, drawing enlargements). Learners who add only vague questions are prompted: "What exactly would you measure?"

Source: Khan Academy (English - US curriculum) Search ARES for similar readings				
Model Building Phase VIDEO: Dilating shapes: shrinking <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Construct scale drawings Source: Khan Academy (English - US curriculum) Search ARES for similar readings	(5 min) Learners open the draft enlargement model started in Lesson 4 ($LSF = \text{image length} / \text{object length}$) and add the new idea: $ASF = k^2$. In their exercise books they draw a revised diagram of S and P labelled with 3, 4, 6, 8, $k = 2$, area 12 cm ² and 48 cm ² , $ASF = 4$ and cost KES 30 and KES 120, with an arrow 'cost follows area, not length'. They add a note: only valid if the figures are similar (both dimensions use the same k), with the distorted 6 x 4 cm photo as a counter-example. They write one 'still unsure' question for the model, such as how volume scales.	Say: "Update your model. What is new today, and how does it change our explanation of Achieng's photo?" Give WAIT TIME of 10-15 seconds, then cold-call 2 learners to describe a new feature on their model. Check that models show LSF and ASF as separate numbers with correct arrows. Ask: "Where does our model still fail to explain something?" and record the gap about volume for Lesson 7. Say: "Keep your model; we will build on it tomorrow."	Cumulative Model Revision: adding ASF to the existing LSF model makes the learning visible and shows how each lesson adds a piece to explaining the photo-resizing phenomenon.	Look for models that (1) keep $LSF = k$, (2) add $ASF = k^2$ with the correct numbers 12 to 48 and 4, (3) link cost KES 30 to KES 120 with area, (4) note the condition that figures must be similar. Learners whose models still show cost doubling to KES 60 get one-to-one feedback before the next lesson.

D. TEACHER REFLECTION

Did learners discover $ASF = k^2$ from their own grid-paper counts, or did they rely on being told? Which learners still believed the cost should be KES 60 after the Explain phase, and what evidence changed (or did not change) their mind? Was the counting of H (1.5 x 2 cm) manageable on the grid, or should half-squares be taught beforehand? How well did learners distinguish LSF from ASF in their words and tables? Which new DQB questions will best lead into Lesson 7 on volume scale factor? What should I adjust in group roles or time so that every learner counts and explains at least once?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We drew prints S (3 x 4 cm), P (6 x 8 cm), L (9 x 12 cm) and H (1.5 x 2 cm) on grid paper and found areas of 12, 48, 108 and 3 cm ² . Four copies of S fit exactly on P, and nine copies fit on L. The prices at KES 2.50 per cm ² were KES 30, 120, 270 and 7.50. The distorted 6 x 4 cm photo had area 24 cm ² .
What did I learn?	When a figure is enlarged with linear scale factor k , its area is multiplied by the area scale factor $ASF = k^2$. For $k = 2$, $ASF = 4$; for $k = 3$, $ASF = 9$; for $k = 1/2$, $ASF = 1/4$. Cost follows area, not length. The rule needs both dimensions to be multiplied by the same k (similar figures).

How does this explain the phenomenon?

Achieng's print P has every length doubled (LSF 2), so its area is $2^2 = 4$ times that of S (12 cm² to 48 cm²). Mr Otieno charges KES 2.50 per cm², so the price rises from KES 30 to KES 120, which is four times and not double. The distorted photo has only the width doubled, so its area (24 cm²) is only twice and it is not similar.

LESSON 7 (40 minutes (double lesson of 80 minutes recommended if cubes are built by hand)): Volume Scale Factor: From Photo Area to Cube and Jerrycan Volume

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners extend the pattern $ASF = k^2$ from Lesson 6 to three dimensions. They use 2 cm cube models and similar jerrycans to discover that volume is multiplied by k^3 when lengths are multiplied by k .
Knowledge	Volume scale factor $VSF = k^3$, where k is the linear scale factor (LSF). A 2 cm cube (8 cm ³) enlarged by LSF 2 gives a 4 cm cube (64 cm ³), and $64 / 8 = 8 = 2^3$. An LSF 3 cube has side 6 cm and volume 216 cm ³ , which is 27 times the original. A 5 L jerrycan enlarged by LSF 2 (height 25 cm to 50 cm) holds $5 \times 8 = 40$ L. 1 L = 1000 cm ³ .
Skills	Building and counting cube models; calculating volume ratios; predicting and testing patterns; using mathematical reasoning and modelling (NGSS SEP: Developing and Using Models; Analyzing and Interpreting Data; Using Mathematics and Computational Thinking; Constructing Explanations).
Attitudes	Curiosity about why growth in three dimensions is faster than growth in length; collaboration in groups; responsibility when handling shared materials; appreciation of similarity and enlargement in day-to-day life (KICD Key Inquiry: How do we apply similarity and enlargement in day-to-day life? Where else in our environment do we see objects that are similar?).
Key Inquiry Question	How do we apply similarity and enlargement in day-to-day life? Where else in our environment do we see objects that are similar? (Lesson focus: how does volume change when an object is enlarged by LSF k ?)
Purpose in Storyline	In Lesson 4 learners calculated $LSF = \text{image} / \text{object}$. In Lesson 6 they showed that $ASF = k^2$ and that Mr Otieno's cost follows area (KES 30 to KES 120 for $k = 2$). This lesson asks what happens when a solid, not a flat photo, is enlarged. It adds $VSF = k^3$ to the cumulative model so that in Lesson 12 learners can explain length, area, cost and volume together for Achieng's passport photo.
Safety Notes	Use sugar cubes, wooden cubes, or paper nets of 2 cm cubes; do not use sharp tools. Keep walkways clear of jerrycans. Use only clean, EMPTY, labelled jerrycans or pictures of them. Do not fill with fuel or chemicals; if water is used, use clean water and wipe spills immediately to prevent slipping.

B. LESSON OVERVIEW

Ms Wanjiru reminds the class of Achieng's prints: Mr Otieno charged KES 30 for print S and KES 120 for print P, four times and not double, because $ASF = 2^2 = 4$. Then she asks: 'What if Mr Otieno sold a solid thing, not a flat photo, such as a jerrycan twice as tall as a small similar one?' Learners predict the volume, build a 2 cm cube from 1 cm cubes and a 4 cm cube (LSF 2), count the cubes (8 and 64), and then check an LSF 3 cube (216 cm³, 27 times). They compare the pattern with the 5 L and 40 L similar jerrycans, state $VSF = k^3$, add it to the class model and update the DQB.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners recall the photo result:	Say: 'Last lesson we learnt that	Predict-Observe-Explain with	Check the prediction cards: do

 VIDEO: Area and Volume of Similar Solids Principles Source: CK-12  Search ARES for similar videos  HTML: Area and Volume of Similar Solids (Flexbook) Source: CK-12  Search ARES for similar readings	print S (3 x 4 cm, KES 30) to print P (6 x 8 cm, KES 120), where every length doubles but the area is multiplied by 4. Ms Wanjiru shows a small similar jerrycan (5 L, height 25 cm) beside a drawing of a larger jerrycan with LSF 2 (height 50 cm). In pairs (3 minutes) learners write on a prediction card: 'If every length doubles, the large jerrycan will hold ____ L because ____.' They also give a prediction for a 2 cm cube enlarged to a 4 cm cube: 'The volume will be ____ cm³.' Likely predictions are 10 L (double), 20 L (like area, x4) or 40 L. Learners record these on the card and keep it for later.	when Achieng's photo doubled in every length, the area became 4 times. Today the photo becomes a solid, a jerrycan. Think quietly first.' Hold up the 25 cm jerrycan (or a picture labelled 5 L). Ask: 'If a jerrycan twice as tall, twice as wide and twice as deep, same shape, is made, how many litres will it hold?' WAIT TIME 15 seconds with no talking, then 1 minute pair talk. Cold-call 4 pairs for their predictions WITHOUT saying whether they are right. Write all predictions on the board (10 L, 20 L, 40 L, others). Ask: 'What was the rule for area? Did it follow the length rule or the length squared rule?' Cold-call 2 learners. Then say: 'Keep your card; we will test it with cubes.'	public prediction recording: committing to a prediction and hearing competing ideas exposes the common misconception that volume doubles when length doubles, and creates a need to test it. Building on the area pattern from Lesson 6 lets learners use an analogy as a hypothesis.	learners give reasons based on the area pattern (e.g., 'area was k^2 so volume may be bigger')? Note how many predicted 10 L, 20 L and 40 L. Look for learners who use 'LSF 2' language and units (L, cm³). Listen in pairs for the misconception 'double length means double volume' and record names to follow up in the Observe phase.
Observe Phase  VIDEO: Area and Volume of Similar Solids Principles Source: CK-12  Search ARES for similar videos  EXERCISE: Area and Volume of Similar Solids Practice Source: CK-12  Search ARES for similar readings	In groups of 4, learners get 64 small 1 cm cubes (or sugar cubes/ paper cubes) and a data table. Task 1: Build a 2 cm cube (side 2 cm) and count the 1 cm cubes: 8 cm³. Task 2: Enlarge it by LSF 2: build a cube of side 4 cm and count the cubes: 64 cm³ ($64 / 8 = 8 = 2^3$). Learners count in layers (each layer $4 \times 4 = 16$, four layers) to check. Task 3: For an LSF 3 cube from the 2 cm cube, work out side 6 cm and volume 216 cm³ using the layers idea ($6 \times 6 \times 6$) without building it, then check that $216 / 8 = 27 = 3^3$. Task 4: Study the jerrycan data: small jerrycan 5 L, height 25 cm; similar jerrycan with LSF 2 has height 50 cm; learners calculate the volume ($5 \times 8 = 40$ L) and convert 5 L and 40 L to cm³ (5000 cm³ and 40,000 cm³; 1 L = 1000 cm³). Learners complete the table: LSF k, side, volume, volume ratio, with a column for k^3. For comparison	Circulate with a checklist. Say: 'Build first, count second, and write the number down before you change anything.' Ask groups: 'How many cubes are in the bottom layer? How many layers?' WAIT TIME 10 seconds before any hints. If a group says 16 cubes for the 4 cm cube (using area thinking), ask: 'Show me the cubes in one layer, then how many layers are there?' Cold-call 3 groups to report the count for the 4 cm cube (64) and 2 groups for the LSF 3 prediction (216). Ask: 'Which two columns look related, side ratio and volume ratio? Where have you seen this pattern before?' Prompt: 'Does the volume ratio follow k, k^2 or k^3?' Collect group results on a class table on the board. Keep jerrycans on a table, never on the floor.	Data table with patterns across rows (Analyzing and Interpreting Data): learners physically build and count to see the 8-fold increase before any formula, then add columns for k, k^2 and k^3 so the pattern can be seen. Layer-counting links the concrete cubes to the algebra ($4 \times 4 \times 4$).	Observable indicators: groups correctly count 8 and 64 cm³; they write $64 / 8 = 8$; they calculate $6 \times 6 \times 6 = 216$ and $216 / 8 = 27$; they convert 5 L and 40 L to cm³ correctly. Note groups who confuse volume with surface area or who say 'x4'. Ask these groups to recount the layers. Record on the checklist how many groups completed the table correctly before whole-class discussion.

	they add the ASF values from Lesson 6 (k^2).			
Explain Phase  VIDEO: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Area and Volume of Similar Solids (Flexbook) Source: CK-12  Search ARES for similar readings	Learners use their tables to write a claim, evidence and reasoning statement: 'When the LSF is k, the volume is multiplied by ____, because ____.' They explain with the cube layers: the length, width and height each get multiplied by k, so volume is multiplied by $k \times k \times k = k^3$. They compare the class results: LSF 2 gives VSF 8 (2 cm cube 8 cm³ to 4 cm cube 64 cm³; jerry can 5 L to 40 L); LSF 3 gives VSF 27 (8 cm³ to 216 cm³). They revisit their prediction cards, correct them in a different colour, and explain why the area rule (k^2) was not enough for solids. Finally, in pairs they solve: 'A similar jerry can has LSF 3 compared with the 5 L jerry can. How many litres does it hold?' ($5 \times 27 = 135$ L) and 'A cube of side 2 cm is reduced by LSF $1/2$. What is its new side and volume?' (1 cm; 1 cm³, VSF = $1/8$). They also link the three results: length $\times k$, area $\times k^2$ and volume $\times k^3$, using the photo: S to P, LSF 2, ASF 4, cost $\times 4$.	Say: 'Using your table, finish this sentence: when every length is multiplied by k, the volume is multiplied by ...' WAIT TIME 15 seconds, then 2 minutes in pairs. Cold-call 3 learners to give the rule and 2 learners to explain it using layers. Push with: 'Why three k's? What are the three directions on the cube?' Ask: 'Why did Mr Otieno's photo only need k^2 and the jerry can needs k^3?' Cold-call 2. Use the misconception Ms Wanjiru heard earlier: 'Someone said 5 L becomes 10 L. What would the cube show?' Ask learners to mark their cards with corrections. Give the 135 L and 1 cm³ problems; WAIT TIME 10 seconds, then check by cold-calling 3 pairs. Also clarify: 'We use the same k for all three directions; if only one direction is changed, as in the distorted 6 x 4 photo, the object is not similar, so VSF = k^3 does not apply.' Do not re-teach the area rule; refer back only to ASF = k^2 from Lesson 6.	Claim-Evidence-Reasoning with prediction revision: learners justify the rule with their cube and jerry can data and algebraic reasoning ($k \times k \times k$), and then explicitly revise their earlier thinking, which shows how evidence changes ideas. Comparing k, k^2 and k^3 side by side builds a generalisation across one, two and three dimensions.	Look for: the correct statement VSF = k^3; evidence from both cubes and jerrycans (8, 64, 216, 5 L to 40 L); correct answers 135 L and 1 cm³; correct use of $1/8$ for reduction. Success criterion: learner can explain WHY it is k^3 (three dimensions) and not just state it. Exit check (1 minute on a slip): 'A cube of side 3 cm has volume 27 cm³. What is the volume with LSF 2, and why?' (216 cm³, 27×8). Note learners who answer 54 and revisit the layers with them.
Driving Question Board (DQB) Creation  VIDEO: Dilating in 3D <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Area and Volume	Learners return to the DQB that Ms Wanjiru started in Lesson 1. In groups they write on sticky notes (a) one question that has now been answered (e.g., 'Why does a bigger jerry can hold more than double?'), moving it to the 'Answered' column with the evidence (8 cm³ to 64 cm³, 5 L to 40 L), and (b) one new question, e.g., 'How is the centre of enlargement found?', 'If Mr Otieno cut a big photo from a smaller one, how do we draw it?', 'How do we use a scale in maps and drawings, and what happens to areas?', 'Do	Say: 'Look at the board. Which of our questions did today's cubes and jerrycans help to answer? Show me the evidence on your sticky note.' WAIT TIME 10 seconds. Cold-call 3 groups to move a question and read the evidence. Say: 'Now write one new question that today made you curious, such as where else we see similar objects around us.' Collect 4 to 5 new questions and cluster them (centre of enlargement, negative and fractional scale factors, maps and costs). Tell learners that Lessons 8	Driving Question Board as a public record of growing understanding: moving questions to 'Answered' with evidence shows progress, and new questions keep learners motivated to continue the investigation of the passport photo phenomenon.	Check that sticky notes cite evidence (8 to 64 cm³, 5 L to 40 L, 216 cm³), not just opinions. Note whether the new questions show curiosity about further parts of the sub-strand (constructing enlargements, scale factors, maps). Count how many groups moved at least one question. Use the new questions to adjust Lessons 8 to 11.

of Similar Solids Practice Source: CK-12 Search ARES for similar readings	similar objects in our environment, such as water tanks in Kisumu, follow the same rule?'. Each group places notes on the DQB and explains one in 20 seconds.	to 10 will answer the questions about construction, and Lesson 11 will answer the questions about maps and costs. Do not answer them now.		
Model Building Phase VIDEO: Using similar & congruent triangles - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili) Search ARES for similar videos EXERCISE: Dilations: scale factor <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners open their class enlargement model, begun in Lesson 4 (LSF = image / object) and extended in Lesson 6 (ASF = k^2 and cost follows area). In groups they add a third row for volume: 'Volume is multiplied by VSF = k^3 '. They complete a summary diagram with one example for each row using Achieng's data: LSF 2 (S to P: 3 to 6 cm, 4 to 8 cm), ASF 4 (12 to 48 cm ² , KES 30 to KES 120), VSF 8 (2 cm cube 8 cm ³ to 4 cm cube 64 cm ³ ; jerrycan 5 L to 40 L). They add a note: LSF 3 gives ASF 9 (12 to 108 cm ² , KES 270) and VSF 27 (8 to 216 cm ³). They also write a caution: the rule only works for similar figures (the distorted 6 x 4 photo is not similar). Each group writes a 2-sentence explanation: 'When a passport photo is enlarged by LSF k , lengths are multiplied by __, area (and so cost) by __, and for a solid, volume by __.' Groups share in a gallery walk.	Say: 'Add today's evidence to our model. What new part does the model need? Use Achieng's photo and the jerrycan.' WAIT TIME 10 seconds. Circulate and ask: 'Where is k in each row? What changes from one row to the next?' Cold-call 3 groups to read their 2-sentence explanation, and ask the class to give thumbs up if it matches the evidence. Ask: 'Which part of the model still cannot explain how the picture is drawn or how to find the centre?' Link to Lessons 8 to 10. Close by asking all learners to write the one-line rule: 'LSF k , ASF k^2 , VSF k^3 '. Remind learners to return all cubes and jerrycans to the table.	Cumulative model revision with a gallery walk: learners extend the model they built earlier by adding a new component (VSF), connect all three scale factors to the same anchoring phenomenon, and identify the limits of the model (similar figures only; the construction of enlargements is still unknown).	Check that models correctly show k , k^2 and k^3 with matching numbers (2, 4, 8; 3, 9, 27) and units (cm, cm ² , cm ³ , KES, L). Look for learners who state the limit that the figures must be similar. Check the 2-sentence explanations for accurate use of 'image', 'object', LSF, ASF and VSF. Record who still mixes up k^2 and k^3 for follow-up in Lesson 8.

D. TEACHER REFLECTION

After the lesson, reflect: (1) How many learners moved from predicting 10 L or 20 L to explaining 40 L with k^3 and the cube layers? (2) Did learners link the cube layers (4 x 4 x 4) to the algebra, or did they only memorise 'cube it'? (3) Which learners still confuse ASF = k^2 with VSF = k^3 , and which representation helped them (cubes, table or photo)? (4) Did every group handle the cubes and jerrycans safely? (5) Did the new DQB questions point to the construction of enlargements (Lessons 8 to 10) and maps and costs (Lesson 11)? (6) What will I change next time: more cubes, a ready-made table, or a pair-check step?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We built a 2 cm cube (8 cm ³) and a 4 cm cube (64 cm ³), compared the 5 L jerrycan with the similar jerrycan that holds 40 L, and calculated an LSF 3 cube of 216 cm ³ (27 times).
What did I learn?	When every length is multiplied by the linear scale factor k , the volume is multiplied by $VSF = k^3$. The pattern is length $\times k$, area $\times k^2$ and volume $\times k^3$, and it works only for similar objects.
How does this explain the phenomenon?	Doubling every length of a solid gives $2 \times 2 \times 2 = 8$ times the volume, because length, width and height each double. That is why a similar jerrycan with LSF 2 holds 40 L, not 10 L or 20 L, and why Achieng's photo (LSF 2) costs 4 times as much because of ASF = 4. This adds volume to the model of her passport photo.

LESSON 8 (40 minutes (one double lesson may be used if available)): Enlargement with a Positive Scale Factor: Locating the Centre and Drawing Rays

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners construct the image of a triangle under an enlargement with a positive scale factor by drawing rays from the centre of enlargement, and explain how this connects to the way print P is an enlargement of print S.
Knowledge	An enlargement needs a centre and a scale factor. Each image point lies on a ray from the centre through the object point, at a distance k times the object point's distance from the centre. For triangle $A(1,1)$, $B(3,1)$, $C(2,3)$, centre $O(0,0)$ and $k = 2$, the image is $A'(2,2)$, $B'(6,2)$, $C'(4,6)$. The image is similar to the object: angles are equal and lengths are multiplied by k .
Skills	Plot points on a coordinate grid; draw rays from a centre through object points with a ruler; measure and mark distances; use the scale factor to locate image points; check the image by comparing corresponding lengths and angles; NGSS-style practice: developing and using models, and constructing explanations from evidence.
Attitudes	Learners show accuracy and care when constructing, appreciate how enlargement is used in photo printing and scale drawing, collaborate with peers (Unity), and take responsibility for rulers and geometry sets (Responsibility).
Key Inquiry Question	1. How do we apply similarity and enlargement in day-to-day life? 2. Where else in our environment do we see objects that are similar?
Purpose in Storyline	Lessons 4-6 gave us the scale factor and showed how the lengths and area of an image relate to the object. This lesson shows WHERE the image goes: the centre and the rays fix its position. It adds the construction part to the class enlargement model so learners can later reverse the process (Lesson 10) and handle fractional and negative factors (Lesson 9).
Safety Notes	Compasses and pencils are sharp: carry them point-down and keep them in the case. Keep rulers flat on desks. Move carefully around the room when sharing work at the board. Bags and chairs must not block walkways.

B. LESSON OVERVIEW

Achieng's photo S became print P with every length doubled, but the shop's photo-resizing software also has to decide where the enlarged picture sits. Learners predict where the image of a small triangle ABC will land when it is enlarged with scale factor 2 from a centre $O(0,0)$. They then construct it by drawing rays from O through A, B and C and marking points at twice the distance, obtaining $A'(2,2)$, $B'(6,2)$, $C'(4,6)$. They measure to check that every side doubles and the angles stay the same, which links back to S and P. They explain the rule in their own words, add questions to the DQB (e.g. what if the centre moves, what if k is less than 1), and add a construction step to the class enlargement model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Dilating shapes:	Learners recall Achieng's prints S (3×4 cm) and P (6×8 cm) and the LSF of 2 from Lessons 4-6. Ms	Display the grid and say: 'Achieng's print P is S with every length doubled. Today we ask	Prediction with reasoning (Predict-Observe-Explain). Writing a prediction before instruction	Observe whether learners (a) say that the lengths of the image will double, (b) mention the centre O

expanding Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Dilations and properties Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Wanjiru gives each pair a grid with triangle ABC at A(1,1), B(3,1), C(2,3) and a marked point O(0,0). Question: 'If we enlarge triangle ABC with centre O and scale factor 2, where will the image land?' Individually, learners sketch a prediction of the image on the grid (a rough triangle) and write one sentence explaining their reasoning. They then compare with a partner. Predictions will vary: some will put the image next to the object, some will double the sides but keep the same corner at A.	where an enlarged picture goes on the page. Predict where triangle A'B'C' will be if the centre is O and $k = 2$. Draw it lightly in pencil.' Give 2 minutes of silent individual work, then WAIT TIME of 10-15 seconds before asking for talk. Cold-call 4 pairs (not volunteers) to describe where they placed A' and why. Do NOT confirm or correct. Record 3 different predictions on the board as Prediction 1, 2, 3. Ask: 'What would you need to know to be sure?' to open up the need for a rule using the centre.	exposes learners' existing ideas, such as 'the image starts at A' or 'just move it over'. This creates a reason to find evidence in the Observe phase and lets learners later see which ideas were supported by their constructions.	as a reference for position, (c) draw the image bigger with the same shape. Tally on a clipboard how many predictions place A' at (2,2), at (1,1) with a bigger shape, or elsewhere. Note learners who give a reason rather than only a drawing.
Observe Phase VIDEO: Dilating triangles: find the error Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Dilation in the Coordinate Plane (Flexbook) Source: CK-12 Search ARES for similar readings	In groups of 3 with a ruler, pencil and grid paper, learners plot A(1,1), B(3,1), C(2,3) and O(0,0). Step 1: draw a ray from O through A, through B and through C, extending each well beyond the triangle. Step 2: measure OA with a ruler (or use coordinates) and mark A' on the ray so that $OA' = 2 \times OA$; repeat for B' and C'. Step 3: join A'B'C' and read the coordinates: A'(2,2), B'(6,2), C'(4,6). Step 4: measure AB, BC, CA and A'B', B'C', C'A' and record them in a table, then compare angles using a corner of paper or a protractor. They find each image length is 2 times the object length and angles are equal. They then compare OA'/OA, OB'/OB and OC'/OC and notice all equal 2. Fast groups also test the ray from O through the midpoint of AB.	Circulate with a checklist. Say: 'Place your ruler at O and the point A. Draw the line past A. Do not stop at A.' Ask: 'How far is A from O? How far should A' be from O?' If a group puts A' on the wrong side or measures from A instead of O, ask: 'Where does the distance start?' WAIT TIME of 10-15 seconds after each question. Stop at 12 minutes and cold-call 3 groups to read their image coordinates; write them on the board and compare. Ask: 'What do you notice about the rays? What do you notice about every image point compared with its object point?' Use the coordinates to show (1,1) to (2,2): both coordinates doubled. Avoid giving the final rule; let learners state it.	Guided construction with data table and patterns. Drawing the rays physically shows that the image points are fixed by the centre and the scale factor, not chosen freely. The table of lengths links the new construction to the earlier idea of LSF, and noticing that the coordinates also double turns a drawing into a general rule.	Look for: rays drawn through O and each object point; A', B', C' marked at twice the distance from O; correct coordinates A'(2,2), B'(6,2), C'(4,6); table showing $AB = 2$ cm and $A'B' = 4$ cm with a ratio of 2 (using the grid unit as 1 cm). Check that learners measure from O. Use a thumbs signal: 'Show me 1, 2 or 3 fingers for how confident you are placing an image point.' Note groups needing support.
Explain Phase VIDEO: Dilating points Source: Khan Academy (English - US curriculum) Search ARES for similar videos	Learners return to their predictions and mark each as supported, partly supported or not supported by the construction. In pairs they write a 'recipe' of 3-4 steps for enlarging any figure with a positive scale factor from a centre. They	Say: 'Turn to your partner. Use your drawing to explain how you know A'B'C' is an enlargement of ABC. Use the words centre, ray and scale factor.' WAIT TIME of 10-15 seconds before pairs speak. Cold-call 3 pairs for their recipe	Claim-Evidence-Reasoning with Think-Pair-Share. Learners claim the image is an enlargement, cite their measured data, and reason with the ray rule. Writing the recipe in their own words consolidates the procedure and surfaces	Listen for correct vocabulary (centre, ray, scale factor, image, object). Check recipes include distance from the centre being multiplied by k. Look for a correct reason that the width-only doubling is not an enlargement (the height

 EXERCISE: Dilation in the Coordinate Plane Practice Source: CK-12  Search ARES for similar readings	then connect the result to the passport photo: 'In print P, which point plays the role of the centre?' and explain that the picture is similar to S because every point is moved along a ray from the centre to k times the distance. They discuss why the image of ABC is similar to ABC, using equal angles and doubled sides, and what the image would look like if only the x-coordinates doubled (like the distorted 6 x 4 photo) and why that is not an enlargement. Several pairs share recipes.	and ask a 4th pair to add one thing that is missing. Ask: 'Mr Otieno doubles only the width of the photo to get 6 x 4 cm. Does that follow the ray rule? Why not?' Ask for evidence: 'Which measurements prove it?' Build the class recipe on the board: 1) mark the centre; 2) draw a ray through each object point; 3) multiply the distance from the centre by k; 4) join the image points and check lengths and angles. Clarify that the centre can be any point, and Lesson 10 will go backwards to find it.	misconceptions, such as measuring from A instead of O or doubling only one coordinate.	did not change, so the shape is stretched and not similar). Collect one exit card: 'Write the image of the point (4,2) with centre O and $k = 2$.' (Answer: (8,4).) Count the number correct.
Driving Question Board (DQB) Creation  VIDEO: Dilating shapes: expanding Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Dilations and properties Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners go to the DQB that Ms Wanjiru has been using since Lesson 1. Each learner writes one new question or one answer on a sticky note, for example 'What if the centre is not O?', 'What if k is less than 1?', 'Can the image be on the other side of the centre?', 'Does the area of A'B'C' follow $ASF = 4$?' They place notes under 'Answered', 'Partly answered' or 'New question'. The class moves the question about the image position to 'Partly answered' and discusses which new questions will be addressed in Lessons 9 and 10.	Say: 'Look at the DQB. What did we learn today that helps answer how a passport photo can be made bigger without changing its shape? Write one new question or one answer on a note.' WAIT TIME of 10-15 seconds. Cold-call 4 learners to read and place their notes and say why. Ask: 'Which questions can we answer from today's construction, and which need more evidence?' Highlight questions pointing to Lesson 9 (fractional and negative k) and Lesson 10 (find the centre). Do not answer the future questions now. Keep the DQB visible for the next lesson.	Driving Question Board as a public record of growing understanding. Sorting questions by status shows learners which parts of the phenomenon they can now explain and keeps the sequence coherent and learner-driven.	Check that notes are specific (mention centre, ray, k) rather than vague. Count how many learners posted a new question about position or the scale factor. Look for learners who connect today's finding to the cost or area question. Record at least 4 questions that show sensemaking.
Model Building Phase  VIDEO: Dilating triangles: find the error Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners revise the class enlargement model started in Lesson 4. Individually, then in groups, they add a new component: 'Position: centre + rays'. On a poster or in their books they draw a diagram of triangle ABC and A'B'C' with the rays from O, labelling $OA' = 2 \times OA$, and next to it a small picture of print S and print P with an arrow labelled 'LSF	Say: 'Open your model from Lesson 4. What new part do we need to add so that it shows where the image goes? Draw it and label it.' Give 5 minutes for work and WAIT TIME of 10-15 seconds before sharing. Cold-call 3 groups to show their updated models and ask: 'What does the model now explain about Achieng's photo that it did not explain before?' Ask:	Cumulative model revision. Adding a new component and listing its limits makes the model an evolving explanation, connecting the construction (centre and rays) to the earlier idea of scale factor and to the photo phenomenon. Naming limits sets purposeful questions for Lessons 9 and 10.	Check that models include: the centre O, rays, image points A', B', C', the statement that distances from O are multiplied by k, and a link to print S and print P. Look for a clear limitation (e.g. 'we have not done k less than 1'). Use a 3-level look-for: Level 1 draws the image only; Level 2 adds rays and the rule; Level 3 adds the link to the photo and a limitation. Record the

 HTML: Dilation in the Coordinate Plane (Flexbook) Source: CK-12 Search ARES for similar readings	2, every length $\times 2$ '. They write a short explanation in one or two sentences: 'An enlargement with scale factor k from centre O moves each point along a ray from O to k times its distance from O . Lengths become k times as long and angles stay the same, so the image is similar.' They also record what the model cannot yet explain: reductions, an inverted image, finding the centre when it is unknown.	'What part of the model is still missing?' Have learners write one limitation. Collect models for use in Lesson 9. Close by saying: 'Next lesson we will see what happens when k is a fraction and when k is negative.'		count at each level.
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D. TEACHER REFLECTION

Which learners measured distances from the centre O rather than from A , and what questions helped them correct this? Did learners' predictions move from 'the image starts at A ' towards the ray rule, and which evidence convinced them? Were the rays drawn accurately enough for the table of lengths to show a ratio of 2? How well did learners link the construction to the photo and to the earlier LSF idea? Which DQB questions should be prioritised in Lessons 9 and 10? What would you change about group roles or the amount of ruler work next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We drew rays from the centre $O(0,0)$ through $A(1,1)$, $B(3,1)$ and $C(2,3)$ and marked each image point at twice its distance from O . We got $A'(2,2)$, $B'(6,2)$, $C'(4,6)$. Each side of the image was 2 times as long as the matching side of the object and the angles were equal.
What did I learn?	An enlargement needs a centre and a scale factor. With a positive scale factor k , each image point lies on the ray from the centre through the object point, at k times the distance from the centre. The image is similar to the object, just like print P is similar to print S .
How does this explain the phenomenon?	Achieng's print P is an enlargement of print S with scale factor 2. Every point of the picture is moved along a ray from the centre to twice the distance, so every length doubles and the angles stay the same, so the shape is unchanged. Doubling only the width (6×4 cm) does not follow this rule, so the photo is stretched and not similar.

LESSON 9 (40 minutes): Shrinking and Flipping: Fractional and Negative Scale Factors

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners construct reductions ($0 < k < 1$) and inverted images ($k < 0$) by drawing rays from the centre of enlargement, and link this to how Mr Otieno can shrink print P back to print S.
Knowledge	An enlargement with $0 < k < 1$ gives a smaller image on the same side of the centre as the object (still called an enlargement with a fractional scale factor). A negative k gives an inverted image on the opposite side of the centre. Image distance from the centre = $ k \times$ object distance. $k = \text{image length} / \text{object length}$. Using $A(1,1)$, $B(3,1)$, $C(2,3)$ and centre $O(0,0)$: $k = 2$ gives $A'(2,2)$, $B'(6,2)$, $C'(4,6)$; $k = 1/2$ applied to $A'B'C'$ returns $A(1,1)$, $B(3,1)$, $C(2,3)$; $k = -2$ gives $A''(-2,-2)$, $B''(-6,-2)$, $C''(-4,-6)$. Print P (6 x 8 cm) to print S (3 x 4 cm) has $k = 1/2$.
Skills	Constructing images on a coordinate grid by drawing rays from the centre; multiplying coordinates by k ; measuring and comparing lengths of image and object; predicting and justifying with evidence; developing and using models (NGSS SEP); analysing and interpreting data (NGSS SEP); communication and collaboration.
Attitudes	Accuracy and care in construction; curiosity about why the image flips; unity when working in groups; responsibility in handling rulers and grid paper; appreciation of enlargement in everyday life such as photo printing.
Key Inquiry Question	How do we apply similarity and enlargement in day-to-day life? Where else in our environment do we see objects that are similar?
Purpose in Storyline	Lesson 8 showed how to construct an image with a positive scale factor by drawing rays from the centre. Lesson 9 extends the idea to reductions and inverted images. It answers the question: how does Mr Otieno go from print P back to print S, and what happens if the scale factor is negative? It prepares Lesson 10, where learners find the centre and scale factor from a given object and image.
Safety Notes	Handle sharp pencils and compass points carefully. Keep rulers flat on the desk. Walk, do not run, when moving to the DQB. Pack away grid paper and tools at the end.

B. LESSON OVERVIEW

Achieng asks Mr Otieno if print P (6 x 8 cm, KES 120) can be made back into print S (3 x 4 cm, KES 30). Learners predict what scale factor does this and what a negative scale factor would do. They then construct on a coordinate grid, with centre $O(0,0)$ and triangle $A(1,1)$, $B(3,1)$, $C(2,3)$: first $k = 2$ gives $A'(2,2)$, $B'(6,2)$, $C'(4,6)$ as a recap from Lesson 8, then $k = 1/2$ applied to $A'B'C'$ returns A , B , C , and finally $k = -2$ gives $A''(-2,-2)$, $B''(-6,-2)$, $C''(-4,-6)$ on the opposite side of the centre and upside down. They measure lengths and compare k with image/object, then connect P to S as $k = 1/2$ and revise the class model and DQB.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Dilations: center Source: Khan	Learners see Achieng holding print P (6 x 8 cm, KES 120) and asking Mr Otieno for print S (3 x 4 cm, KES 30). Individually they write a	Display the sentence: 'Achieng wants print P made smaller to print S. Which scale factor does this?' Say: 'Take 2 minutes silently. Write	Predict-Observe-Explain (POE) with public prediction recording. Committing to a prediction first exposes learners' ideas (for	Look for: learners who write $k = 1/2$ or 'half' for P to S ($6/3 = 8/4 = 2$ so reverse is $1/2$); learners who say a negative factor puts the

Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar readings	prediction on the question: What scale factor takes P to S? Then, if the scale factor were negative, e.g. -2, where would the image of a triangle be: same side as the object or opposite side of the centre? Will it look the same shape? They sketch a quick prediction of the position of the image on a mini grid with centre O and triangle A(1,1), B(3,1), C(2,3), and share with a partner.	your number and one reason.' WAIT TIME: 10-15 seconds after asking: 'What do you think a negative scale factor would do?' Then cold-call 4 learners, asking each: 'What did you predict and why?' Record predictions on the board without judging them. Say: 'We will test every one of these predictions with a construction today.'	example that a negative factor makes a smaller image or that the image stays on the same side) so that the later evidence creates real cognitive conflict and ownership of the reasoning.	image on the opposite side; learners who say shape changes (misconception). Tally on the board how many predict $1/2$, 2, or other, and how many predict same side or opposite side for $k = -2$.
Observe Phase VIDEO: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Dilations: center Source: Khan Academy (English - US curriculum) Search ARES for similar readings	In groups of 3-4, learners use grid paper with axes and centre O(0,0). Step 1 (recap from Lesson 8): plot triangle A(1,1), B(3,1), C(2,3); draw rays from O through each vertex and mark A'(2,2), B'(6,2), C'(4,6) for $k = 2$. Step 2: starting from A'B'C', draw rays from O through each vertex, mark the points halfway along each ray ($k = 1/2$) and read the coordinates: they land on A(1,1), B(3,1), C(2,3). Step 3: draw the rays from O through A, B, C and extend them backwards through O to the opposite side; mark points twice as far from O as the object on this side to get A''(-2,-2), B''(-6,-2), C''(-4,-6). Learners measure AB, A'B' and A''B'' (2 cm, 4 cm, 4 cm) and the height of each triangle, and record the data in a table: object length, image length, $k = \text{image/object}$, side of the centre, orientation (upright or inverted). Each group links the result to print P to S: 6 to 3 and 8 to 4, so $k = 1/2$.	Circulate with the prompts: 'Show me the ray from O through C. How far is C'' from O compared to C?' and 'Which side of O is A'' on?' Say: 'Measure with the ruler first, then calculate.' WAIT TIME: 10-15 seconds after 'What do you notice about the position of A''B''C'' compared with ABC?' Do not give the rule; ask: 'What stays the same, and what changes?' Cold-call 3 groups to report their coordinates for A'', B'', C''. Check that learners write the negative signs. Ask groups that finish early: 'Where would the image be if $k = -1/2$?'	Data table with structured observation (Analysing and interpreting data, NGSS SEP). Recording object length, image length, k and position side by side lets learners see patterns: k controls the size, and the sign controls the side of the centre and the inversion. Constructing with rays gives a physical way to see why distances from the centre scale by k.	Observable indicators: rays are drawn through O and each vertex with a ruler; A'' B'' C'' are correctly plotted at (-2,-2), (-6,-2), (-4,-6); table shows $k = 1/2$ for A'B'C' to ABC, $1/2$ for P to S, and -2 for A to A''; learners state that A''B''C'' is on the opposite side of O and upside down but has the same shape (similar). Note groups plotting A'' on the same side as A, or using $k = 2$ instead of -2, for targeted questioning.
Explain Phase VIDEO: Dilating shapes: shrinking by 1/2 Source: Khan Academy (English - US curriculum) Search ARES	Groups prepare a short explanation using evidence: (1) why $k = 1/2$ makes a reduction, and why image points lie halfway along the rays from O on the same side as the object; (2) why $k = -2$ gives an image on the opposite	Say: 'Use your table and construction as evidence. Start your answer with: Our evidence shows...' Pose: 'Achieng says a smaller photo is a different shape. Is she right?' WAIT TIME: 10-15 seconds, then ask a learner to	Claim-Evidence-Reasoning (CER) with sentence frames and think-pair-share. Learners must connect their claim (reduction shrinks, negative inverts) to the measured evidence and the reasoning that distances from the centre are	Indicators: learners cite specific coordinates or lengths as evidence (e.g. 'C'' is (-4,-6), twice as far as C(2,3) from O and on the opposite side'); they state $k = 1/2$ for P to S using $3/6$ or $4/8$ inverted as $\text{image/object} = 3/6$; they say the

for similar videos EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar readings	side of O, twice as far away, and inverted; (3) why the shape is unchanged in all cases (angles equal, sides in the same ratio of k). Learners complete the sentence frames: 'When k is between 0 and 1, the image is ____ and on the ____ side of the centre.' 'When k is negative, the image is ____ side of the centre and ____.' They then answer: if Mr Otieno reduces print P to print S with $k = 1/2$, what happens to the width (6 to 3), height (8 to 4) and diagonal (10 to 5)? They use their construction data to defend the claim that print S is the same shape as P.	respond and another to add on. Cold-call 4 learners for the explanations of reduction, inversion, shape and the P to S link. Press for precision: 'How do you know the angles are the same? What is the ratio of corresponding sides?' Correct the language: a fractional k is still called an enlargement; say 'object' and 'image'. Do not re-teach finding LSF; refer learners to the ratio image/object they used in Lesson 4.	multiplied by k. This builds mathematical argumentation and a reusable explanation for the phenomenon.	shape stays similar, with the image flipped for negative k. Misconception check: learners claiming a negative k makes the image smaller or changes the shape. Use a quick 3-finger check: 'How many agree that $k = -2$ makes the image bigger than the object?'
Driving Question Board (DQB) Creation VIDEO: Dilations: center Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Dilations: center Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners add new questions and findings to the DQB led by Ms Wanjiru. Examples: 'Why does a negative scale factor flip the picture?' 'Can I find the centre if I only see the object and image?' (leads to Lesson 10) 'If P is reduced to S with $k = 1/2$, does the cost go back from KES 120 to KES 30?' They place a sticky note under the heading 'What we now know' with: '$k = 1/2$ reduces and keeps the image on the same side of O.' '$k = -2$ gives an inverted image on the opposite side.' They review earlier questions and mark any that are now partly answered.	Say: 'Write one question you still have about resizing Achieng's photo on a sticky note.' WAIT TIME: 10-15 seconds before first volunteers. Cold-call 3 learners to read their questions aloud and say where on the DQB each belongs (Lengths, Area and cost, Centre and scale factor). Ask: 'Which earlier question can we now answer with today's evidence?' Highlight the questions that will be addressed in Lesson 10 (finding the centre and k) and Lesson 11 (maps and costs). Keep the DQB name unchanged.	Driving Question Board as a living record of progress. Sorting questions by theme and linking them to evidence shows learners how their understanding grows and creates purpose for the next lesson.	Indicators: questions are relevant to the phenomenon (shape, length, area, cost, centre); at least some questions go beyond today's learning, for example about finding the centre or about cost after reduction; learners can point to evidence from today's construction when placing a note under 'What we now know'.
Model Building Phase VIDEO: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar videos	Learners revise the class enlargement model (first drafted in Lesson 4, built on in Lessons 6-8). They add two new rules in words and diagrams: (1) $0 < k < 1$: image is smaller, on the same side of the centre as the object, distance from the centre multiplied by k (A'B'C' to ABC with $k = 1/2$; print P to S with $k = 1/2$). (2) $k < 0$: image is on the	Say: 'Add today's evidence to our model. What new rule do we need for the centre and the side the image falls on?' WAIT TIME: 10-15 seconds before asking for contributions. Cold-call 3 learners to give one rule each: one for $0 < k < 1$, one for negative k, one for the link to P to S. Ask: 'Does our model still explain why print S is	Cumulative model revision with gallery review. Adding to the model with diagrams and words consolidates the new ideas into one framework and shows how the model now covers enlargements, reductions and inversions.	Indicators: the model sketch shows rays through O, the correct coordinates for A', A'' and A; each rule states both the size and the position of the image; learners can state the k for P to S. Exit ticket: correct answer is 'on the opposite side of O, twice as far from O, and inverted'. Identify learners who still confuse the sign of k with the size

 EXERCISE: Dilations and properties <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	opposite side of the centre, inverted, with size given by k (A to A" with $k = -2$). They draw one labelled sketch with the centre O, triangle ABC, A'B'C' and A"B"C", showing the rays. They complete the summary table below and write one sentence connecting the model to the passport photo: 'Mr Otieno uses $k = 1/2$ to change print P to print S; all lengths halve and the shape stays the same.'	the same shape as P?' and 'What can our model not yet explain?' (how to find the centre; what happens to area and cost when $k = 1/2$, which will be revisited through $ASF = k^2$ from Lesson 6). Collect exit sticky notes with the answer to: 'Where is the image if $k = -2$, and how big is it?'		of the image for follow-up.
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D. TEACHER REFLECTION

Did learners place images for negative k on the opposite side of the centre without reverting to the same side? Which learners confused a fraction (reduction) with a negative number? Did the P to S link help them see that a reduction is still an enlargement with a fractional scale factor? Were measurements accurate enough to show that the image is similar to the object? Which DQB questions show readiness for finding the centre and scale factor in Lesson 10? What would I change in grouping, timing or grid paper use?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We drew rays from the centre O(0,0) through A(1,1), B(3,1), C(2,3). With $k = 2$ we got A'(2,2), B'(6,2), C'(4,6). With $k = 1/2$ applied to A'B'C' we returned to A(1,1), B(3,1), C(2,3). With $k = -2$ we got A"(-2,-2), B"(-6,-2), C"(-4,-6) on the opposite side of O, upside down. Print P (6 x 8 cm) to print S (3 x 4 cm) has $k = 1/2$.
What did I learn?	If $0 < k < 1$, the image is a smaller copy on the same side of the centre (a reduction, still an enlargement with a fractional scale factor). If k is negative, the image is on the opposite side of the centre and inverted, and its size is given by k. In both cases $k = \text{image length} / \text{object length}$ and the shape stays similar.
How does this explain the phenomenon?	Mr Otieno can shrink print P to print S using $k = 1/2$, so every length halves (6 to 3, 8 to 4, diagonal 10 to 5) while the angles stay the same, so the photo is not stretched. Distances from the centre are multiplied by k, which is why negative k flips the image to the opposite side of the centre.

LESSON 10 (40 minutes (one double period may be used if constructions are slow)): Finding the Centre and Scale Factor of an Enlargement from Object and Image

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners work backwards from a finished enlargement. Given an object triangle and its image, they find where the enlargement was made from (the centre) and how big it was (the scale factor k). This extends Lessons 8-9, where they built images from a known centre and scale factor, and it moves the model of how a passport photo is resized on to the question: what do we need to know to describe any enlargement exactly?
Knowledge	Learners will know that: (1) lines drawn through each pair of corresponding points (A and A', B and B', C and C') all meet at one point, the centre of enlargement; (2) the scale factor is $k = \text{image length} / \text{object length}$, using corresponding sides in the same units and order; (3) an enlargement is fully described by its centre and its scale factor. Worked data: object A(2,1), B(4,1), C(3,3); image A'(3,2), B'(7,2), C'(5,6); centre (1,0); $k = A'B'/AB = 4/2 = 2$.
Skills	Learners will: draw straight lines accurately with a ruler through corresponding points on a coordinate grid and extend them until they meet; read coordinates of the intersection point; measure or calculate corresponding lengths to find k ; check the answer by testing a third pair of points (C and C'); explain reasoning with evidence. NGSS-aligned practices: Analysing and Interpreting Data, Developing and Using Models, Constructing Explanations, Engaging in Argument from Evidence.
Attitudes	Learners will show accuracy and patience when drawing, respect for the ideas of peers when comparing constructions, perseverance when lines do not meet exactly, and appreciation that mathematics lets us work out hidden information from what we can see. Values: Unity (group work) and Responsibility (careful use of rulers and shared grids). Competencies: Critical Thinking and Problem Solving; Communication and Collaboration. PCI: Life Skills (self-esteem) through every learner contributing a line or a calculation.
Key Inquiry Question	1. How do we apply similarity and enlargement in day-to-day life? 2. Where else in our environment do we see objects that are similar?
Purpose in Storyline	Lessons 4-7 established the linear scale factor, ASF and VSF. Lessons 8-9 showed how to build images from a centre. Lesson 10 reverses the process: from object and image only, we recover the centre and k . This lets Achieng describe the enlargement from print S to print P exactly (LSF 2) and prepares learners for the maps and scale drawings of Lesson 11 and the final explanation of the passport photo in Lesson 12.
Safety Notes	Low risk. Use rulers and sharpened pencils carefully. Do not point pencils or compasses at others. Keep desks clear so that large grid sheets do not fall. Learners who share sheets should take turns so that everyone draws.

B. LESSON OVERVIEW

Ms Wanjiru recalls Achieng's two prints in Kisumu. Mr Otieno made print P from print S and nobody told us where he scaled from or by how much. Learners first predict how we could find out when we only have the object and the image. They then gather evidence on coordinate grids: they plot triangle ABC with A(2,1), B(4,1), C(3,3) and its image A'(3,2), B'(7,2), C'(5,6), draw lines AA', BB' and CC', and see that all three lines meet at one point (1,0). They measure $AB = 2$ and $A'B' = 4$ and calculate $k = 2$. In the Explain phase they justify why the lines must meet at the centre, check with the distances from the centre, and apply the method to the photo case, where print P is the image of print S with $k = 2$. The DQB is updated with new questions and answers, and the class adds a third section to the enlargement model: how to recover the centre and the scale factor from an object and its image.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners recall Achieng's prints S (3 x 4 cm) and P (6 x 8 cm) from Kisumu and Lesson 8-9 constructions by rays from a centre. Ms Wanjiru shows triangle ABC with A(2,1), B(4,1), C(3,3) and the image A'B'C' with A'(3,2), B'(7,2), C'(5,6) drawn on a board grid, with no centre marked. In pairs, learners write a prediction: 'If we only have the object and the image, where do you think the centre is, and how could we find it? What is the scale factor?' They sketch where they believe the centre lies and state how sure they are (1-5).	Say: 'Mr Otieno gave Achieng print P, but he forgot where he enlarged from and by how much. We only have the object and the image. What can we do?' Remind learners: 'In Lesson 8 and 9 we drew rays from the centre through each point. What if we draw those lines backwards?' Give 2 minutes of silent individual thinking, then pairs share. WAIT TIME: after asking 'Where might the centre be?' wait 10-15 seconds before taking answers. Cold-call 4 pairs and record all predictions on the board without judging. Ask: 'What would you need to measure to find the scale factor?' Wait 10 seconds and cold-call 2 learners.	Prediction with confidence rating (Predict-Observe-Explain). Writing a prediction and a confidence score makes learners' initial ideas visible, creates a reason to test them, and lets them see later whether their thinking has changed. Reasoning backwards from the finished image activates the Lesson 8-9 ray model.	Look for: learners who suggest joining corresponding points; learners who propose comparing lengths of corresponding sides; learners who suggest guessing a point 'in the middle' (misconception). Tally how many of the 4 pairs mention lines through corresponding points. Collect 5 prediction slips and record common ideas and misconceptions to revisit in the Explain phase.
Observe Phase  VIDEO: Dilations: center Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Dilations: center Source: Khan Academy (English - US curriculum) Search ARES for similar readings	In groups of 3-4, learners receive a coordinate grid (x from -2 to 8, y from -2 to 7). Step 1: plot object A(2,1), B(4,1), C(3,3) and join to make triangle ABC. Step 2: plot image A'(3,2), B'(7,2), C'(5,6) and join to make A'B'C'. Step 3: with a ruler draw a long line through A and A' (the line $y = x - 1$), extending in both directions. Step 4: draw a line through B and B'. Step 5: draw a line through C and C'. Step 6: mark and read the point where the lines meet: (1,0). Step 7: measure $AB = 2$ cm and $A'B' = 4$ cm (1 square = 1 cm), also BC and B'C', and calculate $k = A'B'/AB = 4/2 = 2$. Step 8: check using distances from the centre: the centre to A is 1 square across and 1 up, the centre to A' is 2 across and 2 up, so the image is twice as far. Groups record in a table: object length, image length, ratio.	Circulate and say: 'Use a sharp pencil and extend your lines past the triangles. Do your lines have to meet exactly?' Ask: 'Which two lines do you need to find the meeting point? What does the third line do?' If lines miss, say: 'Check the plotting of one point first. Which point could be wrong?' Ask 'What do you notice about the image compared with the object? Which way do the sides face?' (parallel). WAIT TIME: 10-15 seconds after 'What do you notice about the three lines?' before responses. Cold-call 3 groups to state the coordinates where their lines meet and 2 groups to give k with working. Record on the board: 'Centre (1,0), $k = 2$ '. Prompt: 'Is the answer the same for every group? Why?'	Data table and structured observation with the Consensus Check. A table of corresponding lengths and ratios shows that all ratios are equal ($AB:A'B' = 1:2$), connecting to the LSF of Lesson 4. Comparing results across groups builds confidence that the method is reliable, not accidental.	Observable indicators: lines drawn with a ruler through correct pairs (A-A', B-B', C-C'); intersection read as (1,0) within half a square; correct $k = 2$ with the working $4/2$; students note that object and image sides are parallel and angles are equal. Checklist with three levels per group: lines drawn correctly, centre located, k calculated. Note groups that divide object by image (giving $1/2$) to address in the Explain phase.

	They also record what happens to the angles at A and A'.			
Explain Phase VIDEO: Dilating triangles: find the error Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners explain why the three lines meet at one point by using the Lesson 8-9 idea: an enlargement sends every point along a line from the centre, so each image point lies on the line through the centre and its object point; therefore the lines through corresponding points must all pass through the centre. They explain why $k = 2$ by comparing both the side lengths ($A'B' = 4$, $AB = 2$) and the centre distances (centre to A' is twice centre to A). They discuss the common error of dividing object by image and see that this gives $1/2$, the scale factor for going from image back to object (print P to S). Then they connect to the phenomenon: print P (6×8) is the image of print S (3×4) with $k = 2$, so any copy shop can find exactly how the photo was enlarged. In pairs they complete the sentence frames: 'The centre is where ... meet. The scale factor is image length divided by ... length. For S to P, $k = \dots$ '. Extension: learners test by using the centre (1,0) and $k = 2$ on C(3,3): the vector from (1,0) to C is (2,3), doubled it is (4,6), so C' = (5,6), matching.	Ask: 'Why do all three lines meet at one point? Turn and tell your partner.' WAIT TIME 15 seconds, then cold-call 3 learners for the reason. Press for evidence: 'What in your drawing shows that? How does Lesson 8-9 help us?' Say: 'Some groups got $1/2$. What did you divide? Which direction is that scale factor?' Cold-call 2 groups with $1/2$ to explain, and make clear it is the scale factor from image to object. Connect: 'Achieng's print P is the image of print S. What is the scale factor from S to P using widths?' ($6/3 = 2$). Ask: 'If Mr Otieno also made print L (9×12), what k from S?' (3; only the ratio, no re-teaching of area). Then say: 'To describe an enlargement exactly, what two things must we state?' Cold-call 4 learners to complete the statement: centre and scale factor.	Claim-Evidence-Reasoning (CER) with sentence frames. Learners state a claim ('The centre is (1,0)'), support it with evidence (the three lines meet there; distances from the centre double), and give reasoning (an enlargement moves points along lines through the centre). The reverse check with C builds understanding that construction and calculation agree.	Listen for correct reasoning: 'image points lie on lines from the centre'; use of 'corresponding' and 'image/object'; k stated as image over object. Look for: correct statement 'enlargement with centre (1,0) and scale factor 2'; the ability to say S to P has $k = 2$. Use a 1-minute exit question on a slip: 'Object side 3 cm, image side 9 cm. What is k ?' (3). Misconception check: any learner giving $1/3$ should revisit the ratio order.
Driving Question Board (DQB) Creation VIDEO: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar videos	Learners return to the Driving Question Board kept by Ms Wanjiru: 'How can a passport photo be made bigger or smaller without changing its shape, and how do its length, area and cost change when we do so?' Learners add sticky notes in two colours: (a) answers they can now give (for example: 'Print P is an enlargement of S with $k = 2$, and we can find the centre by drawing	Say: 'Take two minutes and write one thing you can now answer and one question that is still puzzling you.' WAIT TIME 10-15 seconds after asking 'Which question should we investigate next?' Cold-call 4 learners to read and place notes and explain where they belong. Point out the link: 'Question about maps will be investigated in Lesson 11.' Ask: 'Does the centre change the size	Driving Question Board with Question Clustering. Sorting questions into answered and open makes progress visible, gives learners ownership of the next steps, and shows how today's finding connects to the whole phenomenon.	Look for: at least one answered note per group that uses 'centre' and 'scale factor' correctly; new questions that go beyond recall (maps, centre position, cost). Count notes placed under Centre and Scale factor (aim for 8 or more). Note learners who write only recall statements and prompt them with 'What would you like to find out next about Achieng's prints?'

 EXERCISE: Dilations: center Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	lines through corresponding points'; 'The scale factor is image length divided by object length'), and (b) new questions (for example: 'Can we use this to read a map of Kisumu to Nairobi?'; 'What if the lines do not meet at one point, is it still an enlargement?'; 'Does the centre change the cost of the photo?'). They cluster notes under headings such as Centre, Scale factor, Cost, Map.	of the photo?' and guide learners to see the centre fixes the position, while k fixes the size. Mark questions answered today with a tick and a date.		
Model Building Phase  VIDEO: Dilations: center Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Similarity & transformations <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners revise the class enlargement model, started in Lesson 4 and extended in Lessons 8-9. In groups they update a poster showing: (1) object and image with labelled points; (2) the centre found by drawing lines through corresponding points (centre (1,0)); (3) the scale factor $k = \text{image length} / \text{object length}$ ($4/2 = 2$); (4) the statement 'An enlargement is described by a centre and a scale factor'; (5) the link to the passport photo: print P is the image of print S with $k = 2$ (3 to 6, 4 to 8, 5 to 10). Groups add an arrow and a note showing the reverse (P to S has $k = 1/2$). They write one sentence about what they can say about the photo's shape (unchanged, equal angles) and give one limitation: 'The model does not yet explain how maps or classroom plans are used' (to be carried forward to Lesson 11).	Say: 'Add to your model what you learned today. Show how we can find the centre and k from an object and its image.' Ask: 'What would a stranger need to know to repeat Mr Otieno's enlargement?' WAIT TIME 10-15 seconds, then cold-call 3 groups to share posters. Ask a challenge: 'If the lines were parallel and never met, what could that tell us?' ($k = 1$ or a translation; accept as a thought to explore, do not teach). Choose 2 posters for a gallery walk and give each group 1 sticky note of feedback: 'One thing clear, one thing to improve.'	Cumulative model revision with a gallery walk. Adding today's method to the existing model links separate ideas (LSF, construction, reverse scale factor) in one picture, and peer feedback pushes learners to justify their model with evidence from the grid.	Rubric-based check: model shows (a) lines through corresponding points meeting at the centre, (b) $k = \text{image/object}$ with the correct values, (c) the link to print S and print P, (d) a stated limitation. Look for correct use of prime notation and units. Collect posters; groups meeting all four criteria are secure, others receive targeted practice in the next lesson.

D. TEACHER REFLECTION

After the lesson ask yourself: Did every learner draw at least one of the three lines with a ruler and read the centre themselves? Which groups divided object by image, and did the Explain discussion clear it up? Did learners reason that image points lie on lines through the centre, or only copy the method? Which learners need more practice with plotting coordinates and extending lines? How well did learners connect the construction back to Achieng's prints S and P? Did the DQB questions show curiosity about maps and cost for Lesson 11? Note which materials (grids, rulers) ran short and adjust the next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Lines drawn through corresponding points A and A', B and B', and C and C' all meet at one point, (1,0). Object AB = 2 and image A'B' = 4. Object and image have equal angles and parallel sides.
What did I learn?	The point where lines through corresponding points meet is the centre of enlargement. The scale factor is $k = \text{image length} / \text{object length} = 4/2 = 2$. An enlargement is described by its centre and its scale factor.
How does this explain the phenomenon?	Print P is an enlargement of print S with $k = 2$ (3 to 6, 4 to 8, 5 to 10), and the shape stays the same because every point moves along a line through the centre and every length is multiplied by the same factor. We can recover the centre and k from only the object and image.

LESSON 11 (40 minutes (double lesson of 80 minutes recommended if the classroom drawing is completed in class)): Maps, Scale Drawings and Costs: Using Scale 1:n in Nairobi, Mombasa and Kisumu

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners apply the idea that a map or scale drawing is a similar figure of the real place, with a scale 1:n acting as the linear scale factor, to find real distances, drawing sizes, areas and costs. They connect this to the passport photo, where cost follows area ($ASF = k^2$) and not length.
Knowledge	Map scale 1:n means actual length = map length $\times$ n. Nairobi to Mombasa: 8.8 cm on the map $\times$ 5,000,000 = 44,000,000 cm = 440 km. Kisumu to Nairobi: 7 cm = 350 km on the same map. Classroom 8 m $\times$ 6 m = 48 m ² drawn at 1:50 is 16 cm $\times$ 12 cm = 192 cm ² ; $ASF = 50^2 = 2500$; $192 \times 2500 = 480,000$ cm ² = 48 m ² . Floor tiles at KES 800 per m ² cost $48 \times 800 =$ KES 38,400.
Skills	Use scale 1:n to convert between map and actual lengths; convert cm to m and km; apply $ASF = k^2$ to scale-drawing areas; calculate cost from area; use mathematical models (NGSS SEP 5: Using Mathematics and Computational Thinking); construct explanations (SEP 6). Core competencies: Critical Thinking and Problem Solving; Communication and Collaboration.
Attitudes	Learners appreciate the use of similarity and enlargement in day-to-day life (maps, building plans, costs); show Unity when working in groups; show Responsibility when handling rulers and shared maps.
Key Inquiry Question	1. How do we apply similarity and enlargement in day-to-day life? 2. Where else in our environment do we see objects that are similar?
Purpose in Storyline	Lessons 4 to 7 gave us LSF, ASF and VSF, and Lesson 5 gave us finding unknown lengths. In Lesson 11 learners test those ideas on larger real-world contexts: a map of Kenya and a classroom plan. This shows that the rule behind Achieng's photos (lengths $\times$ k, area $\times$ k^2 , cost follows area) works beyond photos, preparing for the final explanation in Lesson 12.
Safety Notes	Use rulers and compasses safely; do not point sharp ends at others. Keep walkways clear when learners measure the classroom floor. Share the map and calculators fairly.

B. LESSON OVERVIEW

Ms Wanjiru returns to Achieng's problem: Mr Otieno's Kisumu shop charged KES 30 for print S and KES 120 for print P, which is four times and not double. Learners predict whether the same rules apply when a big place is drawn small. They measure the Nairobi-Mombasa and Kisumu-Nairobi distances on a 1:5,000,000 map, use the 1:50 classroom drawing (16 cm $\times$ 12 cm) to find real size, and calculate tile cost at KES 800 per m². In sensemaking, learners notice that the map scale is a scale factor, that lengths use n while areas use n^2 , and that cost follows area exactly as in the photo prints. The DQB and the class enlargement model are updated.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Scale factors and	Ms Wanjiru displays the prints: S (3 $\times$ 4 cm, KES 30) and P (6 $\times$ 8 cm, KES 120). She then shows a	Say: "Remember Achieng in Kisumu. Print P was twice as wide as print S, yet it cost four times as	Prediction with Explanation (Predict-Observe-Explain): Learners commit to a prediction	Look for: learners who write only a length answer for (b) (misconception: $ASF = n$). Count

area Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Reading Maps, Legends/Scale Practice Source: CK-12 Search ARES for similar readings	classroom plan drawn at 1:50 as 16 cm x 12 cm. Learners individually write a prediction in their notebooks: (a) If the Nairobi-Mombasa road is 8.8 cm on a map with scale 1:5,000,000, how many km is it? (b) The classroom drawing is 16 cm x 12 cm. Is the real floor area 50 times bigger, or more? Why? (c) Will the tile cost follow length or area, as Mr Otieno's paper cost did? Then learners share with a partner and draw a quick sketch to support their idea.	much. Today a map and a classroom plan are our new photos. Predict before we measure." Display the three prompts. Give 3 minutes of silent writing, then WAIT TIME of 10 seconds after asking: "Who thinks the floor is only 50 times bigger in area? Raise your hand." Cold-call 4 learners (different corners of the room) to share predictions without saying whether they are right. Record all ideas on the board. Circulate and note learners who link back to Mr Otieno's pricing rule of KES 2.50 per cm².	and a reason. This surfaces the common misconception that 'area scales the same as length'. The prediction is revisited in the Explain phase so learners see their own thinking change.	how many predict cost follows length versus area. Note learners who mention the photo prints. Record the numbers on a tally chart on the board for comparison in the Explain phase.
Observe Phase VIDEO: Scale drawing: centimeters to kilometers Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Construct scale drawings Source: Khan Academy (English - US curriculum) Search ARES for similar readings	In groups of 4, learners complete three stations with a printed or drawn map of Kenya at scale 1:5,000,000. Station 1 (Nairobi-Mombasa): measure the line with a ruler (8.8 cm), calculate $8.8 \times 5,000,000 = 44,000,000$ cm and convert to 440 km (100 cm = 1 m, 1000 m = 1 km). Station 2 (Kisumu-Nairobi): measure 7 cm, calculate $7 \times 5,000,000 = 35,000,000$ cm = 350 km; compare the two distances. Station 3 (Classroom plan): the plan shows the classroom, actual 8 m x 6 m, drawn at 1:50 as 16 cm x 12 cm. Learners find the drawing area ($16 \times 12 = 192$ cm²), the actual area ($8 \times 6 = 48$ m²), and the tile cost at KES 800 per m² ($48 \times 800 =$ KES 38,400). Groups record results in a table with columns: Item, Map or drawing size, Scale, Real size, Check.	Say: "At each station, write the scale first, then the formula: actual = map x n. Keep the units the same." Give 15 minutes for the stations and rotate roles (measurer, calculator, recorder, checker) after each station. Circulate and ask each group: "What is your linear scale factor here? Which direction are you converting?" Use WAIT TIME of 10 seconds after each question. Cold-call 2 groups to report their Nairobi-Mombasa distance to check that all get 440 km. If a group gets a wrong unit, ask: "How many cm are in 1 km?" and let them find that 1 km = 100,000 cm. Remind them to handle rulers and the shared map carefully.	Data Table and Structured Investigation: Recording evidence in a table makes the pattern visible: every real length is the map length multiplied by n. Rotating roles builds Unity and ensures everyone handles the data.	Check that each group has 440 km, 350 km, 192 cm², 48 m² and KES 38,400 in their table. Observable indicators: measures from the same end points on the ruler; converts 44,000,000 cm to 440 km correctly; writes units on every answer. Note groups that multiply the area 192 cm² by 50 instead of 2500 and use them as a discussion point in the next phase.
Explain Phase VIDEO: Scale factors and area Source: Khan Academy (English)	Learners return to their predictions and compare them with the station data. In their groups they answer: (1) What is the linear scale factor from the map to the real road? From the 1:50 drawing to the real	Say: "Look back at your prediction. Who changed their mind? What evidence changed it?" WAIT TIME of 15 seconds. Cold-call 3 learners who predicted 'area x 50' to explain what they now think. Then	Claim-Evidence-Reasoning (CER) with Talk Moves: Learners must back their claim ('cost follows area, not length') with numbers from the map, plan and prints. Revisiting predictions makes conceptual	Indicators of understanding: learner states that the map scale 1:5,000,000 is a linear scale factor; uses $50^2 = 2500$ for area without prompting; explains that cost uses area ($48 \text{ m}^2 \times \text{KES } 800 = \text{KES}$

- US curriculum) Search ARES for similar videos EXERCISE: Reading Maps, Legends/Scale Practice Source: CK-12 Search ARES for similar readings	classroom? (2) Why did 192 cm² become 48 m² using $ASF = 50^2 = 2500$, giving $192 \times 2500 = 480,000 \text{ cm}^2 = 48 \text{ m}^2$? (3) Why is the tile cost KES 38,400 found from the area (48×800) and not from the length? (4) Link to Achieng: when Mr Otieno doubled the width from print S to print P, the area went from 12 cm² to 48 cm² and the cost from KES 30 to KES 120. Learners write one claim, evidence and reasoning (CER) statement and then share it in a Talk Moves discussion about how maps, plans and photos use the same rule.	pose: "Lengths used 50. Why does area use 2500?" Draw a 1 cm by 1 cm square, then a 50 cm by 50 cm square, and ask learners to count how many small squares fit (2500). Connect it to Lesson 6: "Which rule from the photo prints did we use here?" ($ASF = k^2$). Then ask: "If Mr Otieno charged KES 2.50 per cm² for photos, what is the same about the tile price per m² of KES 800?" Use talk moves: 'Who can add on?', 'Who can say it in a different way?' Correct unit errors (cm² to m²: $10,000 \text{ cm}^2 = 1 \text{ m}^2$) by asking learners to check $480,000 \text{ cm}^2 / 10,000 = 48 \text{ m}^2$.	change explicit and shows that the same scale factor rule works for photos, maps and buildings.	38,400); compares with print P ($KES 120 = 4 \times KES 30$). Collect CER statements as an exit slip and sort them into 'ready', 'nearly' and 'needs support' for follow-up in Lesson 12.
Driving Question Board (DQB) Creation VIDEO: Scale drawing: centimeters to kilometers Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Construct scale drawings Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners revisit the class DQB started in Lesson 1. In pairs they (a) place a sticky note that answers a question already on the board, using evidence from the maps and plans, (b) add one new question that arose today, for example: 'If a map of Kenya at 1:5,000,000 is 8.8 cm for Nairobi-Mombasa, how big would the map area of Kenya be compared with the real area?' or 'What if the scale was 1:2,500,000?' Learners also mark questions that are 'nearly answered' with a tick.	Say: "Which question on our DQB did the map or classroom plan help us answer? Place your note next to it and say why." Give 5 minutes. Cold-call 3 pairs to read a new question aloud, WAIT TIME of 10 seconds before each response. Ask: "Does this question need LSF, ASF or VSF?" Sort questions into categories on the board: lengths, area and cost, volume, centre of enlargement. Keep the board name as Driving Question Board (DQB). Leave 2 or 3 open questions for Lesson 12.	Driving Question Board Tracking: Linking new evidence to earlier questions shows learners that their understanding is growing and gives them ownership of the unit storyline.	Check that each pair has linked at least one note to an existing question and has written a new question. Observable indicator: questions use correct vocabulary (scale factor, ASF, cost, similar). Record questions that still show a length-versus-area confusion.
Model Building Phase VIDEO: Dilations: scale factor <i>Related topic: not an exact match</i>	Learners update the class enlargement model started in Lesson 4 and revised since. On a poster, groups add a new section 'Maps and scale drawings' showing: scale 1:n is a linear scale factor (real = map $\times$ n); area real = drawing area $\times$ n^2 ($192 \text{ cm}^2 \times$	Say: "Add what you learned today to our model. Make sure the rule works for the photo and for the map." Give 8 minutes. Circulate and ask: "Where does the scale factor appear in your model? What happens to the cost?" WAIT TIME of 10 seconds. Cold-call 2 groups	Cumulative Model Revision: Learners revise a shared visual model to include new evidence, making connections across lengths, area, cost and context. Explaining it to Achieng in a sentence checks that the model is communicable.	Indicators: model shows length $\times$ k, area $\times$ k^2 and cost following area; includes the map scale and classroom example with correct numbers; sentence for Achieng states that area becomes 4 times when length doubles so cost goes from KES 30 to KES 120. Note

for this lesson. Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Dilations: scale factor Related topic: not an exact match for this lesson. Source: Khan Academy (English - US curriculum) Search ARES for similar readings	2500 = 480,000 cm² = 48 m²); cost follows area (48 m² x KES 800 = KES 38,400). They add a mini-diagram of the photo prints S to P: lengths x 2, area x 4, cost KES 30 to KES 120, and draw an arrow showing the same rule for the map and classroom plan. Each group writes one sentence to explain to Achieng why Mr Otieno charged more than double.	to present their updates in 1 minute each. Ask the class to give 'a glow and a grow' (one strength and one improvement). Remind them that H (1.5 x 2 cm, KES 7.50) fits the same pattern with $k = 1/2$. Do not re-teach finding centres of enlargement; refer learners to their Lesson 10 poster only if needed.	groups that omit units or mix up n and n^2 and give them a short conference.
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D. TEACHER REFLECTION

Which learners still confuse the length factor n with the area factor n^2 , and what evidence from the tally in the Predict phase shows it? Did the move from cm to m and km cause more difficulty than the scale idea itself? Which group explanation of why cost follows area was clearest, and can it be used to start Lesson 12? How well did learners connect the map and classroom plan back to Achieng's photo prints S and P? What changes would help learners who need more support with unit conversion?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We measured 8.8 cm for Nairobi to Mombasa and 7 cm for Kisumu to Nairobi on a 1:5,000,000 map, and we found that the 1:50 classroom plan is 16 cm x 12 cm (192 cm ²) while the real room is 8 m x 6 m (48 m ²).
What did I learn?	A map scale 1:n is a linear scale factor: real length = map length x n , so 8.8 cm gives 440 km and 7 cm gives 350 km. Area changes by n^2 ($50^2 = 2500$, so 192 cm ² becomes 480,000 cm ² = 48 m ²), and cost follows area (48 x KES 800 = KES 38,400).
How does this explain the phenomenon?	This is the same rule as Achieng's passport photos: when the length is doubled the area is 4 times, so print P costs KES 120 and not KES 60. Maps, plans and photos are similar figures, lengths change by k , areas by k^2 , and cost depends on area.

LESSON 12 (40 minutes): Consolidation: Explaining the Passport Photo

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners pull together everything from Lessons 1-11 to write a final explanation of how Achieng's passport photo can be made bigger or smaller without changing its shape, and how length, area and cost change when this is done.
Knowledge	Enlargement keeps the shape (equal corresponding angles, proportional corresponding sides) and needs a centre and a scale factor. LSF k = image length / object length. ASF = k^2 and VSF = k^3 . Cost at KES 2.50 per cm ² follows area, not length: S (3 x 4 cm, 12 cm ²) costs KES 30, P (6 x 8 cm, 48 cm ²) costs KES 120 and L (9 x 12 cm, 108 cm ²) costs KES 270. A jerrycan with LSF 2 holds 5 L x 8 = 40 L. Doubling the width only (6 x 4 cm) gives a non-similar, stretched photo.
Skills	Constructing a scientific-style explanation (claim, evidence, reasoning) in mathematics; comparing the Lesson 1 model with the final model; using LSF, ASF and VSF to calculate and justify; critiquing a distorted image; and communicating reasoning to peers.
Attitudes	Appreciate the use of similarity and enlargement in day-to-day life (photo shops, maps, models). Show unity when working in groups and responsibility for the group's reasoning. Show confidence in explaining mathematical ideas.
Key Inquiry Question	1. How do we apply similarity and enlargement in day-to-day life? 2. Where else in our environment do we see objects that are similar?
Purpose in Storyline	This is the final lesson. It involves no new content. Learners bring together LSF (Lesson 4), ASF and the cost puzzle (Lesson 6), VSF (Lesson 7) and the idea of the centre and scale factor from object and image (Lesson 10). They answer the DQB and show how their thinking has changed since Lesson 1, when they only predicted that cost would double.
Safety Notes	No hazards. Learners use rulers and calculators carefully and share them fairly. Keep walkways clear during gallery walks.

B. LESSON OVERVIEW

In this closing lesson, Ms Wanjiru returns to Achieng's problem in Kisumu: Mr Otieno prints her photo as S (3 x 4 cm, KES 30) and P (6 x 8 cm, KES 120), and learners must explain why the price is four times and not double. Learners first predict the price and size of print L (9 x 12 cm), then check the prediction with the evidence they already have, using LSF, ASF and cost. They then test a distorted 6 x 4 cm photo and a jerrycan (5 L to 40 L) to show that the same rules apply beyond photos. Learners write a Final Explanation, complete the DQB by sorting the questions into answered and still open, and compare their Lesson 1 model with the final enlargement model. No new content is taught.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Dilations: scale factor Source: Khan Academy (English)	Learners look at the three prints S (3 x 4 cm, KES 30), P (6 x 8 cm, KES 120) and a new print L, whose width is 9 cm and height is not yet given. Working alone for 2 minutes, they predict in their	Display S, P and the distorted photo on the board. Say: "Achieng wants a bigger print, L, that is 9 cm wide. Without calculating yet, what do you think its height, area and price will be? Write your prediction	Prediction with reasoning (Predict-Observe-Explain). Writing predictions before checking them makes learners' existing ideas visible. These include the Lesson 1 misconception that cost doubles	Look for: (1) learners who say the height of L is 12 cm because they scale 4 by 3; (2) learners who predict the price as KES 90 (3 x 30), showing the persisting length-cost misconception; (3) learners

- US curriculum) Search ARES for similar videos EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar readings	books: (a) the height of L, (b) its area, and (c) its price at KES 2.50 per cm². They also predict whether a 6 x 4 cm photo is an enlargement of S. They then share with a partner and agree or disagree with reasons. Predictions link directly to Achieng's photo problem in Kisumu.	and one reason." Give WAIT TIME of 10-15 seconds of silent thinking before writing. Cold-call 4 learners to give predictions and record them on the board without judging. Ask: "Who predicted the price would be 3 times KES 30? Who thought something else? Why?" Say: "We will test these predictions in the next phase using what we found in Lessons 4, 6 and 7."	when width doubles. They have a reason to test those ideas against evidence.	who justify with ratios or scale factors rather than just 'it looks bigger'. Tally on the board how many predict KES 90 and how many predict KES 270, to decide how much time to spend on the cost reasoning in the next phase.
Observe Phase VIDEO: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Dilations: scale factor Source: Khan Academy (English - US curriculum) Search ARES for similar readings	In groups of 4, learners complete an evidence table for S, P, L and the distorted photo. For each print they record width, height, diagonal (5, 10, 15 cm for S, P, L), area (12, 48, 108 cm²) and price (KES 30, 120, 270). They calculate LSF from S to P ($6/3 = 8/4 = 10/5 = 2$) and from S to L (3), the ASF (4 and 9) and the cost ratios ($120/30 = 4$ and $270/30 = 9$). They then measure the distorted 6 x 4 cm photo: width ratio 2 and height ratio 1, area 24 cm². Finally they use the jerrycan data: a small jerrycan holds 5 L and is 25 cm high; a similar jerrycan with LSF 2 is 50 cm high and holds $5 \times 8 = 40$ L. Each group notes which pattern (k, k^2 or k^3) matches length, area or cost, and volume.	Circulate with a checklist. Ask: "Which two corresponding sides are you dividing? Is the order the same on top and bottom?" Ask: "For S to L, what is the area ratio, and how does it relate to $k = 3$?" Pause for WAIT TIME of 10-15 seconds before accepting an answer. For groups that expect cost to double, say: "Look at your table. Does cost follow the width column or the area column? Prove it with one row." Ask the distorted photo group: "Why is the width ratio 2 but the height ratio 1? What does that tell us about similarity?" Cold-call 3 groups to report which row of the table surprised them.	Data table with pattern finding. Putting all the evidence from Lessons 4, 6 and 7 into one table lets learners see that length scales by k, area and cost by k^2, and volume by k^3, so the separate lessons become a single connected idea.	Check the completed tables for correct values: LSF 2 and 3, ASF 4 and 9, cost ratios 4 and 9, area of the distorted photo 24 cm². Observable indicators: learners write k^2 next to area and cost and k^3 next to the 5 L to 40 L jerrycan; learners note that the distorted photo has unequal ratios (2 and 1) and so is not similar. Record groups that still divide the wrong corresponding sides and re-teach briefly at their table.
Explain Phase VIDEO: Dilations: center <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Dilations: center <i>Related topic: not</i>	Individually, learners write a Final Explanation answering: 'How can a passport photo be made bigger or smaller without changing its shape, and how do its length, area and cost change when we do so?' They must use the CER structure: Claim (P is an enlargement of S with LSF 2), Evidence (3 to 6, 4 to 8, 5 to 10 cm; 12 to 48 cm²; KES 30 to KES 120; equal angles), and Reasoning ($ASF = k^2 = 4$, so cost, which is KES 2.50 per cm², is multiplied by 4 and not 2). They	Model one sentence stem only: "Because every length is multiplied by k, the area is multiplied by ____, so the cost..." Say: "Write as if you are explaining to Achieng why Mr Otieno charges KES 120 and not KES 60." Give WAIT TIME of 10-15 seconds before writing begins. During writing, ask individuals: "Where is your evidence? Which numbers prove it?" After peer review, cold-call 3 learners to read one sentence they strengthened. Ask: "Where else in our	Claim-Evidence-Reasoning with peer critique. Writing a full explanation makes learners link observation (numbers), principle (k, k^2, k^3) and conclusion (cost is four times, not double). Peer review sharpens the explanation and exposes gaps in reasoning.	Use a 4-point rubric: (1) claim stated with LSF 2; (2) evidence includes at least three of: lengths, angles, area 12 to 48 cm², cost KES 30 to KES 120; (3) reasoning uses $ASF = k^2 = 4$ and links cost to area; (4) addresses the distorted photo or the jerrycan ($VSF = 8$). A learner scoring 3 or 4 is secure. Collect explanations from 8 learners to see who still writes 'cost doubles' and plan a follow-up.

<i>an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	add one sentence on why the 6 x 4 cm photo is not similar and one on the jerrycan ($VSF = 2^3 = 8$, 5 L to 40 L). They then swap with a partner and use a 3-item checklist to review: claim, numbers, reason.	environment do we see similar objects, such as maps, a model matatu at 1:25 (28 cm) or shadows in Kericho?" Take 2 further answers from learners.		
Driving Question Board (DQB) Creation  VIDEO: Dilating shapes: shrinking <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Dilations and properties <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners return to the Driving Question Board started in Lesson 1. In groups, they take each question card from the DQB (for example, 'Why is the price more than double?' and 'Is every bigger photo an enlargement?') and decide whether it is Answered, Partly answered, or Still open. For each answered card they write the answer and the lesson that gave the evidence (for example, Lesson 6 for ASF and cost, Lesson 7 for volume). Then they post one new question that extends the idea, such as how reductions or negative scale factors apply to other items. They connect the answered cards to the Driving Question with string or arrows.	Stand by the DQB and say: "Ms Wanjiru's board started with our questions about Achieng's photo. Which of these can we now answer with evidence?" Hand each group 3 cards. Give WAIT TIME of 10-15 seconds before groups confer. Cold-call 5 groups, each to read one card and give its answer with the evidence. Ask: "Which question is still open? What would we need to find out?" Make sure at least one card links to Lesson 10, finding the centre and scale factor from object and image, and one to Lesson 11 (maps and costs). Close with: "Which two of our questions show how we use similarity and enlargement in day-to-day life?"	Driving Question Board sorting. Revisiting the learners' own questions shows how their understanding has grown and keeps the unit tied to the phenomenon. Naming the lesson that supplied the evidence builds the habit of justifying answers with data.	Check that each answer card includes both an answer and a piece of evidence (for example, 'cost x 4 because $ASF = 2^2 = 4$; 12 cm² to 48 cm²'). Observable indicators: learners correctly reject cards based on the old idea that cost doubles; learners generate a new extension question. Note cards sorted wrongly and discuss them with the whole class.
Model Building Phase  VIDEO: Dilating triangles: find the error <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos	Learners take out the first enlargement model they drew in Lesson 1 (before instruction) and place it beside a blank page. Individually they draw the final model: photo S (3 x 4 cm) and P (6 x 8 cm) with a labelled $LSF = 2$, arrows to show corresponding sides, equal angles marked, a box for $ASF = k^2 = 4$ linked to price KES 30 to KES 120, and a box for $VSF = k^3 = 8$ linked to the jerrycan 5 L to 40 L. They add a small note on the centre of	Say: "Put your first model next to your final one. In one colour, show what you got right at the start. In another, show what you changed." Give WAIT TIME of 10-15 seconds before drawing. Circulate and ask: "Where does your model show that the shape stays the same? Where does it show that the price is four times and not double?" Cold-call 3 learners to explain one change from Lesson 1 to the final model. End with an exit question on a slip: "Print L is 9 x 12 cm. Write its LSF	Model comparison (Lesson 1 vs final). Placing the first model beside the final one makes the learners' conceptual change visible and lets them see which ideas they changed, which builds metacognition and ownership of the learning.	Check final models for: $LSF = 2$ labelled with corresponding sides; equal angles shown; $ASF = k^2 = 4$ linked to cost KES 30 to KES 120; $VSF = k^3 = 8$ linked to the jerrycan 5 L to 40 L; at least one marked change from Lesson 1. Exit slip: correct answers are $k = 3$, $ASF = 9$, KES 270 (108×2.50). Learners who answer KES 90 need individual follow-up. Aim for at least 80% correct and record the results.

 HTML: Corresponding Parts of Similar Figures (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	enlargement and the scale factor. They then use two colours to mark what changed from their Lesson 1 model, for example, 'cost doubles' replaced by 'cost $\times k^2$'. Finally, they share the model in a 2-minute gallery walk with a partner.	from S, its ASF, and its price." ($k = 3$, $ASF = 9$, KES 270). Collect slips at the door. Close by saying: "Achieng can now tell Mr Otieno exactly why his charge is four times, and so can you."		
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D. TEACHER REFLECTION

After the lesson, ask yourself: (1) How many learners still wrote that cost doubles when width doubles, and in which phase did their thinking change? (2) Did learners use the data from S, P and L to justify k , k^2 and k^3 , or did they quote the rules without evidence? (3) Could they explain why the 6 x 4 cm photo is not similar? (4) Which DQB questions remain open, and how can I use them to begin the next unit? (5) Did all learners, including those who are quieter, get a chance to speak through the cold-calls and peer review? Note changes to make next time, such as more time for the Final Explanation or a clearer sentence stem.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that print P (6 x 8 cm) is an enlargement of print S (3 x 4 cm): every length is doubled (3 to 6, 4 to 8, diagonal 5 to 10 cm) and the angles stay the same. The area went from 12 cm ² to 48 cm ² and the price from KES 30 to KES 120. Print L (9 x 12 cm) has area 108 cm ² and costs KES 270. The 6 x 4 cm photo, where only the width was doubled, looked stretched. A similar jerrycan with LSF 2 holds 40 L instead of 5 L.
What did I learn?	We learned that an enlargement keeps the same shape when corresponding angles are equal and corresponding sides are in the same ratio. $LSF\ k = \text{image length} / \text{object length}$. $ASF = k^2$ and $VSF = k^3$. Cost at KES 2.50 per cm ² follows area, so it changes by k^2 , not k . We also learned that a negative or fractional scale factor still gives an enlargement, and that a centre and a scale factor can be found from an object and its image.
How does this explain the phenomenon?	Achieng's photo can be made bigger or smaller without changing its shape by multiplying every length by the same linear scale factor. From S to P, $k = 2$, so lengths double, but area is multiplied by $2^2 = 4$ (12 cm ² to 48 cm ²). Mr Otieno charges KES 2.50 per cm ² , so the price rises from KES 30 to KES 120, which is four times and not double. For S to L, $k = 3$, $ASF = 9$ and the price is KES 270. If only the width is doubled (6 x 4 cm), the sides are not in the same ratio, so the photo is not similar. For solids, volume is multiplied by k^3 , so the jerrycan goes from 5 L to 40 L.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.